



**COMSATS University Islamabad,
Sahiwal Campus, Pakistan**

Buy Best

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COMSATS University Islamabad, Sahiwal Campus**

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of the requirement for the degree of**

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “Buy Best” was developed by **Anum Fatima (CIIT/FA22-BCS-020)** and **M Shahwaiz Shahid (CIIT/FA22-BCS-049)** under the supervision of **Bilal Shabbir Qaisar** and co supervisor **Ali Shahid** and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

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Executive Summary

Online shopping has become a common practice, but users often face difficulties when trying to find the best product at the best price. To compare a single item, they must visit multiple e-commerce websites, search repeatedly, and manually check prices, features, and reviews. This process is time-consuming and sometimes confusing, and users may miss better deals simply because they did not explore every platform. Existing comparison tools mainly focus on price and do not provide detailed filtering based on specific product requirements.

To overcome these issues, **Buy Best** is developed as a web-based meta search engine that brings multiple online shopping platforms together in one place. Instead of searching on different websites separately, users can enter their desired product along with specific requirements such as brand, price range, RAM, storage, or other features. The system uses web scraping techniques to collect product data from various e-commerce websites and displays the best matching results in a single, organized interface. This allows users to easily compare products based on specifications, cost, and brand without switching between platforms.

Buy Best enhances the user's experience by offering **intelligent recommendations** based on search behavior and preferences. Users can also save or shortlist products for later review. When they decide to make a purchase, the system redirects them to the original website to complete the transaction securely. Our project provides a smart and efficient solution for online shoppers. By combining product search, comparison, filtering, and recommendation features into one system, it saves time, reduces effort, and helps users make better and more informed purchasing decisions.

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Abbreviations

SRS	Software Require Specification
PC	Personal Computer
API	Application Programming Interface
SDD	Software Design Description
AI	Artificial Intelligence
SQL	Structured Query Language
UI	User Interface
DB	Data Base
FYP	Final Year Project
HTML	Hyper Text Markup Language
OOP	Object Oriented Programming
UML	Unified Modeling Language
SDLC	Software Development Life Cycle
HCI	Human Computer Interaction

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Chapter 1

1. Introduction

This chapter provides an overview of the **BuyBest** project, explaining its background, purpose, and overall objectives. It describes the problem faced by online shoppers, the motivation behind developing the system, and how the proposed solution aims to improve the product search and comparison process.

With the rapid growth of e-commerce, online shopping has become a preferred choice for many people [5]. However, customers often face difficulty when trying to compare products across multiple websites. They must visit different platforms individually, search for the same product repeatedly, and manually compare prices, specifications, and reviews. This process is time-consuming and may lead to confusion [19] or poor purchasing decisions due to incomplete information.

BuyBest is designed as a web-based meta search engine [17][6] that gathers product information from various e-commerce websites and displays it in a single, organized interface. The system allows users to search for products based on specific requirements such as brand, price range, and technical specifications. By using web scraping and intelligent filtering techniques, our project helps users find the most suitable products quickly and efficiently.

The main objective of this project is to simplify online shopping by providing a centralized platform for product search, comparison, and recommendation. Through advanced filtering options and smart suggestions, the system aims to save time, reduce effort, and support users in making informed and confident purchasing decisions.

1.1 Brief

BuyBest is a web-based meta search engine developed to simplify the online shopping experience. The project aims to solve the problem of searching and comparing products across multiple e-commerce websites. Instead of visiting different platforms separately, users can search for a product once and view comparison results, specifications, and prices in a single interface. The outcome of this project is a functional system that gathers product data, applies filtering based on user requirements, and provides intelligent recommendations to support better purchasing decisions.

The system is developed using modern web technologies. The front end is designed to provide a user-friendly interface, while the backend is implemented using Laravel. Web

scraping techniques are used to collect product data from various e-commerce websites, and MySQL is used for structured data storage. An **AI-based recommendation module** is integrated to provide smart suggestions based on user preferences and search behavior.

The development of **BuyBest** follows a systematic software development methodology, including requirement gathering, system design, implementation, testing, and deployment. Data was collected through both primary methods of user surveys and feedback and secondary methods analysis of existing e-commerce and comparison systems. The report discusses the problem statement, literature review, system architecture, tools and technologies, implementation details, testing procedures, and final outcomes of the project. Each chapter highlights a specific aspect of the system, providing a complete understanding of the design and functionality of Buy Best.

1.2 Key Features

The **BuyBest** system provides several important features that improve the online shopping experience for users. These features help users easily search, analyze, and compare products from different e-commerce platforms in one place.

1.2.1 Product Search

One of the main features of the system is product search. Users can search for any product by entering its name, brand, or specific specifications in the search bar. The system processes the query and retrieves relevant product information [6] from multiple e-commerce websites through web scraping . The results are then displayed in a structured and user-friendly interface so that users can quickly view available options.

1.2.2 Product Comparison

Another important feature of the system is product comparison. Users can select multiple products and compare them side by side [19]. The comparison feature displays differences in price, specifications, ratings, and other important attributes. This helps users easily identify which product offers better value and performance.

1.2.3 Advanced Product Filtering

The system also provides advanced filtering options that allow users to narrow down search results based on their preferences. Users can apply filters such as brand, price range, RAM, storage capacity, ratings, and other specifications. This feature helps users

quickly find products that match their exact requirements without browsing through many irrelevant results.

1.2.4 AI-Based Recommendation System

A unique feature of **BuyBest** is the AI-based recommendation system. This module analyzes product specifications and user behavior to suggest similar or better products that the user might be interested in. By using intelligent recommendation techniques, the system helps users discover alternatives that they may not have considered before.

1.2.5 Save or Bookmark Products

Users can also save products for later viewing. The saved products feature allows users to create a personalized list of items that they are interested in purchasing. This makes it easier for users to revisit products and compare them again before making a final decision.

1.2.6 Admin Management Panel

Finally, the system provides an admin management panel where administrators can monitor system activity and manage important system components. The admin can manage users, control data sources used for web scraping, and monitor system performance. This feature ensures that the system operates efficiently and that product data remains updated and accurate.

1.3 Purpose of the Project

The main purpose of our project is to develop a platform that helps users easily search and compare products across multiple online shopping websites [5][17].

Online shoppers often spend a significant amount of time browsing different websites to check product prices, features, and availability. This process can be inefficient and sometimes leads to poor purchasing decisions because users may not have access to all available options.

Our project aims to solve this problem by providing a centralized platform where users can search for products and instantly view results collected from different e-commerce websites. By presenting product information in a structured and organized manner, the system allows users to quickly evaluate different options and select the best product according to their requirements. Another important purpose of the system is to enhance user experience through intelligent recommendations [18]. The **AI recommendation**

module analyzes product features and user behavior to suggest products that match the user's preferences.

Our project is designed to make online shopping faster, easier, and more efficient by reducing the effort required to search for and compare products.

1.4 Scope of the System

The scope of Our project focuses on simplifying the product search and comparison process for online shoppers. With the rapid growth of e-commerce platforms, users often face difficulty when trying to compare products across different websites. Each website presents product information in a different format, which makes it time-consuming for users to evaluate multiple options. The **BuyBest** system addresses this problem by providing a centralized platform where users can search for products and view aggregated results from multiple online stores [17][10].

The system primarily targets users who want to compare electronic products such as smartphones and laptops across multiple e-commerce platforms. These products usually have many technical specifications such as RAM, storage, processor, display size, battery capacity, and price. By collecting and organizing this information in one place, the system allows users to easily analyze different products and choose the most suitable options.

The project does not function as an online shopping store where products are directly sold. Instead, it acts as a meta search engine that gathers product information from different websites and displays it in a unified and structured format. When users select a product, they wish to purchase, they are redirected to the original website where the product is available. In this way, this project serves as an information and comparison platform rather than a direct sales platform.

The system integrates web scraping techniques to collect product data such as specifications, prices, images, and ratings from different e-commerce websites. This information is then stored and organized in a database so that users can quickly access it through the search interface. The scraping mechanism can be updated or expanded to include additional sources when required.

Another important aspect within the scope of the system is the integration of **Artificial Intelligence** based recommendation features. The AI module analyzes product specifications and user search behavior to recommend relevant products. This helps users discover better alternatives based on their preferences and browsing patterns.

The system also provides features that improve user convenience and interaction. Registered users can create accounts, log in securely, and save products that they want to review later. This bookmarking feature helps users keep track of products they are interested in without repeating the search process.

In addition to user functionality, the system includes an administrative module that allows administrators to monitor system operations. Administrators can manage user accounts, control data sources used for web scraping, and monitor analytics related to user searches and system performance.

The system covers the following main functions:

- Product search and filtering
- Product comparison
- AI-based product recommendations
- Product bookmarking or saved products
- User authentication and login
- Admin management and analytics
- Data collection through web scraping
- Product information display from multiple sources

The **BuyBest** system is designed to operate as a web-based application accessible through modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox. The platform provides a responsive user interface so that users can access the system easily from desktops or laptops.

Although the current implementation focuses on electronic products such as smartphones and laptops, the system architecture is designed in a flexible way so that additional product categories can be included in the future. This allows the platform to expand its functionality and support a wider range of e-commerce product comparisons.

Overall, the scope of the Buy Best system is to provide an efficient, user-friendly, and intelligent platform that simplifies product discovery and comparison across multiple online shopping websites.

1.5 Limitations of the System

Although the **BuyBest** system provides several useful features that simplify the online shopping and product comparison process, there are certain limitations associated with the system. These limitations mainly arise due to technical constraints, dependency on external websites, and the scope of the current implementation.

1.5.1 Dependency on Web Scraping

One of the major limitations of the system is its dependency on web scraping to collect product data from external e-commerce websites [2]. The system extracts product information such as price, specifications, and images by analyzing the structure of those websites. If the layout or HTML structure of these websites' changes, the scraping scripts may stop working correctly. In such cases, the scripts must be updated or modified to continue collecting accurate data.

1.5.2 Limited Product Categories

Another limitation of the system is that it currently focuses mainly on specific product categories such as mobile phones and laptops. These categories were selected because they have well-defined technical specifications that can be easily compared. However, expanding the system to include additional product categories such as home appliances, clothing, or accessories may require further development and modification of the data collection and comparison modules.

1.5.3 Accuracy of AI Recommendations

The AI-based recommendation system depends heavily on the quality and availability of product data as well as the user's interaction history [3][18]. If the system has limited user data or incomplete product information, the recommendations may not always be perfectly accurate. The effectiveness of the recommendation engine improves overtime as more user interactions and product data become available.

1.5.4 Dependency on Internet Connectivity

Since the **BuyBest** platform operates as a web-based application, it requires a stable internet connection to function properly. Users cannot access the system or retrieve product information if they are offline or experiencing poor internet connectivity.

1.5.6 Data Update Frequency

Another limitation is related to data updates. Product prices and specifications on e-commerce websites may change frequently. If the scraping process is not performed regularly, the information displayed on the **BuyBest** platform may become outdated, which can affect the accuracy of price comparisons.

Despite these limitations, the Buy Best system still provides a valuable and efficient solution for simplifying the online shopping process by bringing together product search, comparison, and recommendations on a single platform.

1.6 Relevance to Course Modules

Our project is strongly related to various courses studied during the BCS program, as it combines theoretical knowledge with practical implementation. It integrates multiple disciplines including software engineering, web development, databases, artificial intelligence, and human-computer interaction. The project demonstrates how academic concepts are applied to solve real-world problems like product searching and comparison. It also reflects system design, development, and testing practices learned during the degree. Overall, it serves as a complete practical application of core computer science subjects.

1.6.1 Web Engineering and Software Engineering

This project applies key concepts of software engineering such as requirement analysis, system design, UML diagrams, implementation, testing, and deployment [4][9][8]. The development process follows structured methodologies to ensure system reliability and maintainability. Web engineering concepts are used in designing a responsive and interactive web-based platform. The project reflects real-world software development lifecycle practices.

1.6.2 Database Systems

Database concepts are used in designing and managing structured storage using **MySQL**. It includes normalization, relational schema design, and efficient query processing. The system stores product data, user information, bookmarks, and alerts in an organized manner. This ensures fast retrieval and data consistency across the application.

1.6.3 Artificial Intelligence

AI is used in the recommendation module of the **BuyBest** system. It analyzes user behavior, search history, and preferences to generate personalized suggestions. This improves decision-making for users by recommending relevant products. It demonstrates the practical application of AI in e-commerce systems.

1.6.4 Data Structures and Algorithms

Efficient data handling is achieved using sorting and filtering algorithms. Product data is organized to enable fast searching and comparison. The system ensures optimized performance when handling large datasets. This reflects key concepts of algorithmic efficiency.

1.6.5 Programming Fundamentals and OOP

The system is developed using object-oriented programming principles such as modularity, encapsulation, and reusability. Logical programming concepts are used to structure the application efficiently. Code is divided into modules for better maintenance and scalability. This ensures clean and organized system design.

1.6.6 Human-Computer Interaction (HCI)

HCI principles [12] are applied in designing a user-friendly and responsive interface. The system focuses on simplicity, ease of navigation, and clear layout. Users can easily perform actions such as search, compare, and save products. This improves overall user experience.

1.6.7 Project Management

Project management concepts are used for planning, scheduling, and documentation. Tasks are divided into phases such as analysis, design, implementation, and testing. Proper time management ensures timely completion of the project. This reflects real-world project development practices.

1.6.8 Software Testing and Quality Assurance

Testing concepts are applied through unit testing, integration testing, functional testing, and automated testing. Each module is verified for correctness and performance. Bugs are identified and fixed during development stages. This ensures system reliability and quality.

1.6.9 Web Scraping and Data Integration

The system uses web scraping techniques to collect product data from multiple e-commerce platforms. APIs are also integrated to fetch real-time information. Data is cleaned, processed, and standardized before use. This ensures accurate and updated product listings.

1.6.10 System Design and Architecture

System design principles are used to structure the application into layers such as frontend, backend, database, and integration layer. This improves scalability and maintainability. Each layer has a specific responsibility. This ensures smooth system operation and easy future enhancements.

1.7 Project Background

The idea behind **BuyBest** originates from the rapid growth of e-commerce and the increasing dependence of users on online shopping platforms. Today, thousands of products are available on different websites, each offering different prices, specifications, and deals. While this provides more choices for customers, it also creates confusion and difficulty when comparing products across multiple platforms. Users often need to visit several websites separately to check prices, features, and reviews before making a purchase decision.

Our project is based on the concept of a **meta search engine**. This is a system that collects data from multiple sources and displays the results in one unified interface. Instead of storing all products directly like a traditional e-commerce website, it gathers information from different online stores and presents it in a structured and comparable format. This allows users to search once and view results from various platforms simultaneously.

The project also utilizes web scraping, which is a technique used to extract data from websites automatically. The scraped data is processed, organized, and stored in a database so that users can filter and compare products easily. Additionally, the system incorporates intelligent recommendation techniques to suggest products based on user preferences and search behavior.

In simple terms, our project combines the concepts of meta searching, data extraction, comparison systems, and recommendation systems to create a centralized platform for smarter online shopping. The main idea is to reduce user effort, save time, and improve decision-making by bringing multiple e-commerce platforms together into one efficient system.

1.8 Literature Review

With the rapid growth of e-commerce, online shopping has become an essential part of modern life. Many platforms such as Amazon, eBay, and Daraz provide a wide range of

products to customers [5][6]. Although these platforms offer convenience, users still face difficulty when comparing the same product across multiple websites. They need to visit each site separately, check prices, review specifications, and analyze customer feedback manually. This process is time-consuming and may result in incomplete comparisons or poor purchasing decisions.

To address this issue, several price comparison websites have been developed, such as Google Shopping, PriceGrabber, and Shopzilla [19]. These systems collect product prices from different sources and display them in one place. While they reduce the effort required to check multiple websites, most of them mainly focus on price comparison and provide limited filtering based on detailed product specifications. They often lack personal recommendations, local market coverage, and advanced comparison features, which are important for better decision-making.

Research in the field of e-commerce systems highlights the importance of meta search engines that gather data from multiple platforms and present unified results. A **meta search engine** does not sell products directly but collects information from various sources and displays it in a single interface. This approach improves transparency and saves time for users. Web scraping technology plays a major role in such systems by automatically extracting product information such as prices, features, ratings, and availability from different websites. The collected data is then cleaned, organized, and stored in structured databases to allow efficient searching and comparison.

Another important development in online shopping systems is the use of recommendation techniques. Modern platforms apply artificial intelligence methods such as content-based filtering and collaborative filtering to suggest products based on user interests and behavior. These techniques increase user engagement and help customers discover suitable products more easily. Studies show that personalized recommendations significantly improve customer satisfaction and confidence in online purchasing.

In addition, researchers emphasize the importance of advanced filtering and sorting mechanisms. Customers often want to filter products by multiple criteria such as brand, price range, storage capacity, RAM, ratings, and other specifications. Systems that support multi-criteria filtering reduce information overload and make comparison more effective. User interface design is also considered a critical factor, as a simple and well-structured layout enhances usability and improves overall experience.

Despite these advancements, many existing solutions still provide only partial support. Some focus mainly on price, while others concentrate on recommendations without offering proper comparison tools. There is still a need for a unified platform that integrates data extraction, detailed filtering, product comparison, and intelligent suggestions in one system. This project is developed to address this gap. It combines web scraping, structured data management, comparison features, and recommendation techniques into a single platform, aligning with current research trends and contributing to the improvement of online shopping systems.

Recent trends in e-commerce highlight the growing importance of data accuracy, real-time updates, and localized product availability. Customers expect up-to-date pricing information, reliable product details, and access to both international and local sellers. This has increased the demand for systems that can continuously update product data and present trustworthy, organized information. Integrating automated data collection with structured storage and intelligent processing ensures that users receive relevant and current results, further enhancing the effectiveness of modern product comparison platforms like **BuyBest**.

1.9 Analysis from Literature Review

The literature review highlights that existing e-commerce platforms and price comparison [19] systems provide partial solutions to the problem of online product search and comparison. Most traditional e-commerce websites focus only on displaying their own products, which limits users to a single platform. On the other hand, price comparison tools gather data from multiple sources but mainly emphasize price and often lack detailed specification-based filtering, intelligent recommendations, and localized coverage. This creates a gap between user expectations and the services currently available.

In comparison to the systems discussed in the literature, **BuyBest** aims to provide a more integrated and user-centered solution. Unlike individual e-commerce platforms, it does not restrict users to one website. Instead, it functions as a meta search engine that gathers product information from multiple sources and presents it in a unified interface. This approach directly addresses the issue of repetitive searching and manual comparison highlighted in previous studies.

Furthermore, while many existing comparison tools focus primarily on price, our project emphasizes multi-criteria filtering, allowing users to search products based on specifications such as brand, price range, RAM, storage, and other features. This aligns

with research that stresses the importance of advanced filtering mechanisms to reduce information overload and improve decision-making.

The literature also discusses the growing importance of recommendation systems in enhancing user experience. In response to this trend, our project integrates intelligent recommendation features that suggest relevant products based on user preferences and search behavior. This makes the system not only a comparison tool but also a supportive decision-making platform.

Overall, the analysis shows that this project builds upon existing research and current market solutions while addressing their limitations. By combining web scraping, structured data management, advanced filtering, comparison features, and intelligent recommendations in one system, the project offers a more comprehensive and efficient solution for online shoppers.

1.10 Methodology and Software Lifecycle for This Project

The development of the **BuyBest** system follows a structured software development approach to ensure that the project is completed in an organized and efficient manner. The methodology used for this project involves several stages including requirement gathering, system design, implementation, testing, and deployment. Each stage is carefully planned so that the system meets user requirements and performs effectively.

For this project, a **Waterfall Software Development Life Cycle (SDLC)** model [4][11] is selected. This model follows a step-by-step process where each phase is completed before moving to the next one. The first phase involved gathering requirements by studying existing online shopping platforms and understanding the problems users face when comparing products across different websites. This information helped define the main objectives and features of the Buy Best system.

In the second phase, the system design was created. This included designing the overall architecture, database structure, and user interface. Diagrams such as class diagrams, sequence diagrams, and data flow diagrams were prepared to clearly represent how the system components interact with each other.

The next phase involved the implementation and development of the system. In this stage, the frontend interface was developed for user interaction, while the backend was implemented using Laravel to manage application logic and database operations. Web scraping techniques were used to collect product data from multiple e-commerce websites,

and the information was stored in a MySQL database [14]. An AI-based recommendation component was also integrated to suggest relevant products to users.

After development, the testing phase was carried out to ensure that all system features work correctly. Different types of testing were performed to identify and fix errors, improve system performance, and ensure smooth user interaction. Finally, the system was deployed and evaluated to confirm that it meets the project objectives and provides a useful solution for online product search and comparison.

1.11 Rationale behind Selected Methodology

The **Waterfall** SDLC model was selected for our project because the system requirements were clearly defined at the beginning. This model follows a step-by-step process including requirement analysis, design, development, testing, and deployment. Using this structured approach helped maintain proper documentation and ensured each phase was completed before moving to the next.

During implementation, the project follows an Object-Oriented development approach. This approach organizes the system into objects and classes representing real-world entities such as users, products, and search operations. It improves code reusability, maintainability, and scalability, making the system easier to manage and update in the future.

1.11.1 Selection of Waterfall SDLC Model

The **Waterfall** SDLC model was selected for our project because the system requirements were clearly defined at the initial stage. This model is suitable for projects where changes are minimal and objectives are well understood. It allows the development team to follow a fixed path from start to end. As a result, the project becomes easier to plan, manage, and execute.

1.11.2 Reasons for Choosing Waterfall Model

The Waterfall model was chosen because it is simple, easy to understand, and well-structured. Each phase has specific tasks and deliverables, which reduces confusion during development. It ensures that the team focuses on one stage at a time without overlapping with work. This makes the development process more organized and efficient.

1.11.3 Structured Development Approach

The Waterfall model follows a step-by-step approach that includes requirement analysis, system design, development, testing, and deployment. Each phase is clearly defined and completed before moving forward. This structured flow helps in better planning and execution of tasks. It also ensures that the system is developed in a logical and systematic manner.

1.11.4 Importance of Proper Documentation

One of the key advantages of this model is proper documentation at every stage of development. All requirements, designs, and testing results are recorded in detail. This documentation helps in understanding the system and tracking progress. It is also useful for future maintenance, updates, and troubleshooting.

1.11.5 Phase-wise Development Process

The Waterfall model ensures that each phase is completed before moving to the next phase. This helps in identifying errors early and fixing them before they affect the whole system. It also improves the quality of the final product. The step-by-step process makes the development more controlled and easier to manage.

1.11.6 Use of Object-Oriented Development Approach

During the implementation phase, the project follows an **OOP** approach [9]. This approach divides the system into smaller parts called objects and classes. Each object performs a specific function, making the system more organized. It also helps developers manage complex systems more easily.

1.11.7 Representation of Real-World Entities (Objects & Classes)

In OOP, real-world entities such as users, products, and search operations are represented as objects. These objects contain data and functions related to them. This makes the system design more realistic and easier to understand. It also improves the clarity of the code structure.

1.11.8 Advantages of OOP (Reusability, Maintainability, Scalability)

The Object-Oriented approach provides many benefits such as reusability, maintainability, and scalability. Code reusability allows developers to use the same code in multiple parts

of the system. Maintainability makes it easier to fix bugs and update features. Scalability ensures that new features can be added without affecting the existing system.

1.11.9 Future System Maintenance and Upgradability

By using Waterfall and OOP together, the system becomes easier to maintain and upgrade in the future. Developers can easily understand the system through proper documentation and structured code. New modules or features can be added without major changes. This makes the system flexible and long-lasting.

Chapter 2

2. Problem Definition

With the rapid growth of online shopping, customers often face difficulty in finding the best product at the most reasonable price. Different e-commerce websites offer the same products with different prices, specifications, and customer reviews. Because of this variation, users often become confused when deciding which product is the most suitable for their needs. Many customers also struggle to identify whether they are getting the best value for their money. Another major problem is that users have to visit multiple online stores individually to search for the same product and compare its details. This process requires significant time and effort because customers must repeatedly search for the product, check its features, and analyze prices on different platforms. Sometimes important information may be missed during this process, which can lead to purchasing a product that is not the best option available.

The lack of a unified comparison platform makes it difficult for users to make quick and informed decisions. Customers prefer a simple system where they can view product information, prices, and features from different sources in one place. Without such a system, the shopping experience becomes complicated and inefficient.

Our project is designed to address these issues by providing a centralized platform that gathers product information from various online sources and presents it in an organized manner. The system allows users to search for products, compare prices and specifications, and identify the best available option easily. The outcome of this project is a system that saves users time, simplifies the comparison process, and helps customers make smarter and more confident purchasing decisions.

2.1 Problem Statement

Online shopping has become very popular, but users often face difficulties when trying to find the best product at the best price. Different e-commerce platforms offer the same products with different prices, specifications, and reviews. As a result, customers have to visit multiple websites to search for a single product and compare its details, which is time-consuming and sometimes confusing.

The main problem is the absence of a single platform where users can easily search, view, and compare products from different online stores in one place. Because of this, customers may miss better deals or more suitable products available on other websites.

Our project aims to solve this problem by providing a centralized system that gathers product information from multiple online sources and displays it in an organized and easy to understand format. This will help users quickly compare products and make better purchasing decisions.

2.2 Outcome of the Project

The outcome of our project is the development of a web-based **meta search engine** that helps users search, analyze, and compare products from multiple e-commerce websites in one place. The system simplifies the process of finding the best product by collecting product information from different online platforms and presenting it in a structured and organized format.

One of the main outcomes of the project is an integrated search platform that allows users to enter a product name or keyword and receive results gathered from several websites. Instead of visiting multiple online stores separately, users can view different product options through a single interface. This reduces the time and effort required to compare products and find the best deal.

Another important outcome of the project is the implementation of a product comparison feature. Users are able to select multiple products and compare their specifications, prices, and ratings side by side. This comparison functionality allows users to clearly understand the differences between products and make better purchasing decisions.

The system also provides filtering options that allow users to refine search results based on different criteria such as brand, price range, RAM, storage capacity, and other specifications. These filtering capabilities help users quickly find products that match their specific requirements.

An additional outcome of the project is the integration of an **AI-based recommendation system**. The recommendation module analyzes product specifications and user interactions to suggest relevant products. This feature improves the overall user experience by helping users discover alternatives that may better suit their needs.

The project also produces a user-friendly interface that makes the platform easy to use and accessible through modern web browsers. Users can navigate the system, view product details, and perform comparisons without needing advanced technical knowledge.

From a technical perspective, the project demonstrates the use of web technologies, data collection through web scraping, database management, and the integration of artificial intelligence techniques. These components work together to create a functional system that supports efficient product discovery and comparison.

Overall, the outcome of our project is a complete system that improves the online shopping experience by providing a centralized platform for product search, comparison, and intelligent recommendations.

2.3 Deliverable and Development Requirements

The main delivery of our project is a functional web-based application that allows users to search and compare products from different online sources. The system provides product details such as price, specifications, and other important information in a single platform. It also includes features such as product search, comparison of multiple products, and display of organized results that help users choose the best option. In addition, the project deliverables include the complete project documentation, system design, source code, and testing results.

The development of this project requires both software tools and technical resources. The system is developed using modern web technologies including programming languages for front end and back-end development, a database for storing product and user related data, and web scraping tools to collect product information from different websites. Other development requirements include a computer system with internet access, a suitable development environment, and testing tools to ensure the system works correctly and efficiently. These resources help in successfully designing, implementing, and deploying the **BuyBest** system.

2.3.1 Deliverables

Our project produces several deliverables that represent the final outputs of the system development process. These deliverables include both technical components and documentation that demonstrate the design, development, and implementation of the system.

The primary delivery of the project is the **BuyBest** web application. This application provides users with a platform where they can search for products, compare different options, and view product details collected from multiple e-commerce websites. The

application includes features such as product search, filtering, comparison, bookmarking, and AI-based recommendations.

Another important delivery is the web scraping module that collects product data from various e-commerce websites. This module extracts product specifications, prices, images, and ratings and stores them in the system database so that users can access them through the search interface.

The project also delivers an AI-based recommendation component that analyzes product specifications and user interactions to suggest relevant products. This module helps users discover alternatives and improves the overall decision-making process.

Documentation is another key delivery of the project. This includes the Software Requirements Specification document, Software Design Description document, and the Final Year Project report. These documents explain the system requirements, design architecture, implementation details, and testing procedures used in the development process.

In addition, the project provides an administrative panel that allows administrators to manage system operations, monitor user activity, and control the data sources used for collecting product information.

Overall, the deliverables of the **BuyBest** project include the complete working system, supporting modules, and detailed project documentation.

2.3.2 Development Requirements

The development requirements describe the tools, technologies, and resources needed to build the **BuyBest** system. These requirements support the development process and ensure that the application can be properly designed, implemented, and tested.

The development of the Buy Best system requires a suitable development environment that supports web application programming and database management. A code editor or integrated development environment is needed to write and manage the project code. Tools such as Visual Studio Code or similar development environments are commonly used because they provide useful features for coding, debugging, and project management.

Version control systems are also useful during development to manage changes in the project code. These tools allow developers to track updates, maintain different versions of the code, and collaborate effectively during the development process.

The development process also requires web technologies for building the user interface and backend functionality. Frontend technologies are used to design the layout and interaction of the system, while backend technologies handle data processing, database communication, and system logic.

Testing tools are also part of the development requirements. These tools help developers verify that the system functions correctly and that each feature operates as expected.

Development requirements ensure that the project can be built in a structured environment using appropriate tools and technologies.

2.3.3 Software Requirements

The software requirements define the software tools and platforms required to run and support the **BuyBest** system. These software components allow the system to function properly and provide the necessary environment for executing the application.

The system requires a web browser such as **Google Chrome**, **Mozilla Firefox**, or **Microsoft Edge** [5] to access and use the application. These browsers allow users to interact with the system interface and perform product searches and comparisons.

A web server environment is required to host the application and manage user requests. The backend server processes search queries, communicates with the database, and retrieves product information collected from e-commerce websites.

A database management system is also required to store product data, user information, and system records. The database organizes and manages the information so that it can be easily accessed when users conduct searches or comparisons.

Programming languages and frameworks are also part of the software requirements. These tools are used to build the system logic, implement the scraping functionality, and develop the AI recommendation component.

Additionally, libraries and tools used for data collection and machine learning support the implementation of the recommendation system and sentiment analysis features.

Overall, the software requirements provide the digital environment necessary for the development and operation of the Buy Best system.

2.3.4 Hardware Requirements

The hardware requirements define the physical devices needed to develop and operate the **BuyBest** system. These requirements ensure that the system can run efficiently during development, testing, and deployment.

The development of the system requires a computer or laptop with sufficient processing power to support programming tools, databases, and testing environments. A system with a modern processor and adequate memory allows developers to run multiple applications such as code editors, servers, and browsers simultaneously.

Adequate storage capacity is also necessary to store project files, scraped product data, and system databases. The storage device may be a hard disk drive or a solid-state drive that provides reliable and fast access to stored data.

A stable internet connection is required for several purposes, including accessing e-commerce websites for data collection, downloading development tools, and testing the system functionality online.

Input and output devices such as a keyboard, mouse, and display monitor are also essential for interacting with the development environment and operating the system.

Overall, the hardware requirements for the **BuyBest** system are moderate and can be fulfilled using standard personal computers or laptops commonly used for software development.

2.3 Current System

Existing online product search systems like Amazon, eBay, and Walmart require users to visit each platform separately to find product information, compare prices, and check availability, reviews, and recommendations without integrating data from multiple sources or offering AI-based personalization see Figure 2.1.

Our Project addresses these limitations by aggregating product information from multiple e-commerce sites, enabling price and specification comparisons, AI-driven recommendations, and a centralized interface for advanced filtering, side-by-side product comparison, and bookmarking.

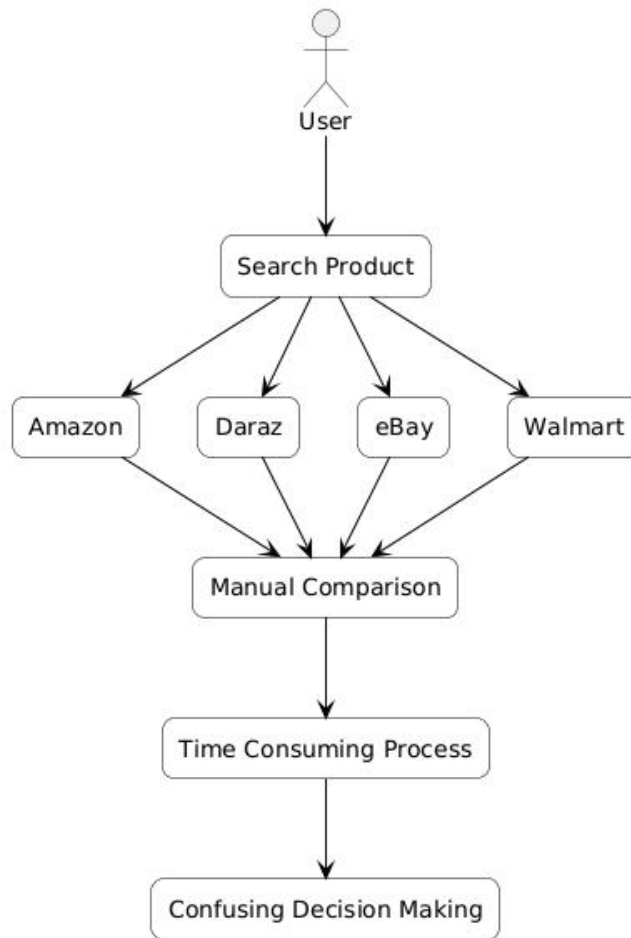


Figure 2.1 Current Product Search System

Table 2.1 Data Dictionary

Feature	Platform Support	Limitations
Product Listings	Amazon, eBay, Walmart	Limited to each platform, no aggregation
Price Comparison	Individual	Users must manually compare across platforms
Filters (Price, Brand, Specs)	Yes, per platform	No cross-platform filtering
User Reviews & Ratings	Yes	Only within one platform
Recommendations	Basic AI/Rule-based	Not personalized across platforms
Product Bookmarking	Yes	Limited to a single platform
Cross-Platform Price Alerts	No	Users must monitor each platform separately

Chapter 3

3. Requirement Analysis

The Requirement Analysis identifies and defines the functionalities and constraints of the Buy Best system. It includes case diagrams and detailed descriptions to show how users and administrators interact with the system, along with functional requirements that specify what the system must do and non-functional requirements that describe performance, security, usability, and reliability standards.

3.1 Use Case Diagram

The Figure of our project illustrates interactions of two actors: User and Admin. Users can search for products, sign up, log in, compare prices, bookmark items, view AI recommendations, and receive price alerts, while Admin manages users, produces reports to control and maintain the system.

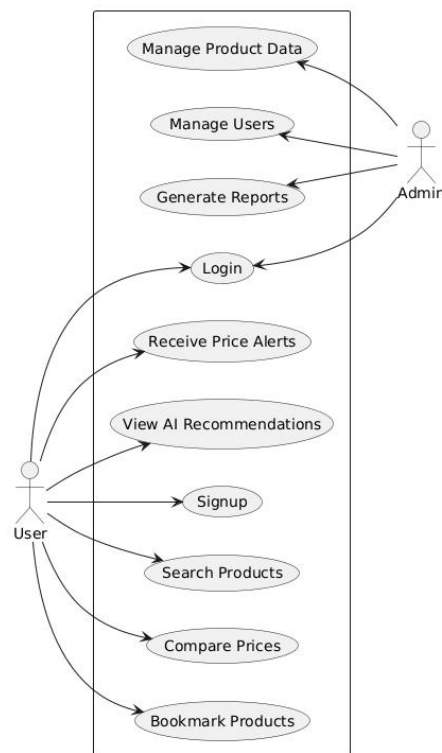


Figure 3.1 Use Case Diagram

3.2 Detailed Use Case

A detailed use case provides a step-by-step description of how a user or administrator interacts with the system to achieve a specific goal. It includes the actor, preconditions, trigger, main flow, alternate flows, and postconditions, helping to clearly define system behavior and ensure all functional requirements are captured.

Table 3.1 Data Dictionary

Use Case ID	UC-1
Actor	User.
Description	Allow users to search for products across multiple e-commerce platforms and compare their prices and specifications.
Preconditions	User is logged into the Buy Best system.
Trigger	Users enter a product name or category in the search bar.
Main Flow	1. User enters search term. 2. System fetches products from multiple sites. 3. System displays product listings with price and specs. 4. User selects products to compare. 5. System displays side-by-side comparison.
Alternate Flow	No results found → System shows a message and suggests similar products. User cancels search → Return to main page.
Postconditions	User views a comparison of selected products and can bookmark or receive alerts.

Table 1.2 Login

Use Case ID	UC-2
Actor	User / Admin
Description	Allow registered users to log into the Buy Best system securely.
Preconditions	Users must have a registered account and internet is available.
Trigger	User clicks the Login button and enters credentials.
Main Flow	1. User enters email and password. 2. System validates credentials. 3. System grants access.

	4. User is redirected to dashboard/homepage.
Alternate Flow	Invalid credentials: System shows error message. User forgets password: Redirect to reset password page.
Postconditions	User is successfully logged into the system.

Table 3.3 Signup

Use Case ID	UC-3
Actor	New User
Description	Allows new users to create an account in the Buy Best system.
Preconditions	Users are not already registered and internet is available.
Trigger	User clicks the Signup button.
Main Flow	1. User enters name, email, password and details. 2. System validates entered data. 3. System creates new accounts. 4. Confirmation message is displayed
Alternate Flow	Email already exists: System asks user to login. Invalid data: System asks user to correct fields.
Postconditions	New user account is created successfully.

Table 3.4 Search Products

Use Case ID	UC-2
Actor	User
Description	Allows users to search for products across multiple e-commerce platforms.
Preconditions	Users are logged into the system and internet is available.
Trigger	User enters a product name or keyword in the search bar.
Main Flow	1. User enters search term. 2. System processes the query. 3. System retrieves product data from database/sources. 4. System displays list of products.
Alternate Flow	No results found: System displays message and suggestions. User cancels search: Return to homepage.
Postconditions	Relevant product results are displayed to the user.

Table 3.5 Compare Prices

Use Case ID	UC-3
Actor	User
Description	Allow users to compare prices and specifications of multiple products.
Preconditions	Search results must be available and at least two products selected.
Trigger	User selects products and clicks compare button.
Main Flow	1. User selects multiple products. 2. System retrieves product details. 3. System displays comparison table with prices and specs.
Alternate Flow	Less than two products selected: System shows warning message.
Postconditions	Comparison of selected products is displayed.

Table 3.6 Bookmark Products

Use Case ID	UC-4
Actor	Users
Description	Allows users to save products for future reference.
Preconditions	User must be logged in.
Trigger	User clicks on “Save” or “Bookmark” option.
Main Flow	1. User selects product. 2. User clicks save option. 3. System stores product in user’s saved list.
Alternate Flow	User not logged in: System prompts login.
Postconditions	Product is saved in user’s account.

Table 3.7 Receive Price Alerts

Use Case ID	UC-5
Actor	Users
Description	Allow users to receive alerts when product price changes.
Preconditions	Users must be logged in and product selected.
Trigger	User enables price alert option.
Main Flow	1. User selects a product. 2. User enables alert. 3. System monitors price changes.

	4. System notifies user when price changes.
Alternate Flow	Notification failure: System retries sending alert.
Postconditions	User receives notification of price changes.

Table 3.8 View AI Recommendations

Use Case ID	UC-6
Actors	Users
Description	Provides users with AI-based product recommendations.
Preconditions	User activity or product data must be available.
Trigger	User views recommendations section or searches products.
Main Flow	1. System analyzes user behavior and product data. 2. System generates recommendations. 3. Recommended products are displayed.
Alternate Flow	Insufficient data: System shows general recommendations.
Postconditions	User views recommended products.

Table 3.9 Manage Product Data

Use Case ID	UC-7
Actors	Admin
Description	Allows admin to manage product data and sources.
Preconditions	Admin must be logged in.
Trigger	Admin accesses product management panel.
Main Flow	1. Admin views product data. 2. Admin updates or deletes product info. 3. System saves changes.
Alternate Flow	Invalid data entered: System shows error.
Postconditions	Product data is updated in system.

Table 3.10 Manage Users

Use Case ID	UC-8
Actors	Admin
Description	Allows admin to manage user accounts.
Preconditions	Admin must be logged in.
Trigger	Admin opens user management section.
Main Flow	1. Admin views users list. 2. Admin adds/updates/deletes users. 3. System saves changes.

Alternate Flow	Invalid input: Error message displayed.
Postconditions	User records are updated.

Table 3.11 Generate Reports

Use Case ID	UC-9
Actors	Admin
Description	Allows admin to generate reports based on system data.
Preconditions	Admin must be logged in and data available.
Trigger	Admin selects report generation option.
Main Flow	1. Admin selects report type. 2. System processes data. 3. System generates report. 4. Report is displayed or downloaded.
Alternate Flow	No data available: System shows message.
Postcondition	Reports are generated successfully.

3.3 Functional Requirements

The functional requirements of the **BuyBest** system define the core features and operations that the system must perform to meet user and administrative needs.

3.3.1 Product Search Across Multiple Platforms

The system allows users to search for products across multiple e-commerce platforms using a single search interface. Users can enter product names, keywords, or specifications, and the system retrieves relevant results from different sources. This feature eliminates the need to visit multiple websites separately and saves time for users.

3.3.2 Display of Product Listings

The system shall display product listings in a structured and user-friendly format. Each product listing includes important details such as price, specifications, ratings, and images. This allows users to quickly review product information and make informed decisions without navigating to external websites.

3.3.3 Product Comparison

The system shall allow users to compare multiple products side by side. Users can select two or more products and view their differences in terms of price, specifications, and ratings. This

feature helps users analyze options easily and choose the most suitable product based on their requirements.

3.3.4 AI-Driven Product Recommendations

The system should provide AI-driven product recommendations based on user behavior and preferences. The recommendation module analyzes user searches, browsing patterns, and product features to suggest relevant or alternative products. This enhances user experience and helps users discover better options.

3.3.5 Product Bookmarking

The system allows users to bookmark or save products for future reference. Users can create a personalized list of saved items, which they can revisit later without performing the search again. This feature is useful for users who want to compare products over time before making a purchase decision.

3.3.6 Cross-Platform Price Alerts

The system should send cross-platform price alerts to users. When a product's price changes on any of the connected e-commerce platforms, the system notifies the user. This feature helps users purchase products at the best possible price and stay updated with market changes.

3.3.7 Admin Management of System Data

The system shall allow administrators to manage product data and user accounts. Admins can add, update, or delete product information, manage data sources used for web scraping, and control user accounts. This ensures that the system remains accurate, secure, and up to date.

3.3.8 Report Generation for Admins

The system shall generate reports for administrators based on system data and user activity. These reports may include user behavior, search trends, and system performance. This feature helps administrators analyze system usage and make improvements where necessary.

3.4 Non-Functional Requirements

- **Performance:** The system shall fetch search results from multiple platforms within 5 seconds. This ensures users experience minimal delay when searching for products. Optimized algorithms and caching techniques will be used to maintain consistent speed even under high load.

- **Scalability:** The system should support up to 10,000 concurrent users. The architecture will allow horizontal scaling by adding servers or cloud instances. Database and server resources will be optimized to handle increasing user demand without performance degradation.
- **Usability:** The system should have a user-friendly interface suitable for desktop and mobile devices. Navigation will be intuitive, with clear menus and easily accessible features. Responsive design principles will ensure all functionalities are usable across different screen sizes and devices.
- **Security:** The system should ensure secure login, data encryption, and prevent unauthorized access. User credentials and sensitive data will be encrypted using industry-standard protocols. Regular security audits and updates will protect against vulnerabilities and cyber threats.
- **Reliability:** The system shall have 99% up time and handle network failures gracefully. Backup and failover mechanisms will ensure minimal disruption in service. Error handling and monitoring tools will detect and resolve issues quickly.
- **Maintainability:** The system shall allow easy updates for new e-commerce platforms and features. Modular code design will simplify adding or modifying functionalities. Documentation and automated testing will support faster maintenance and troubleshooting.

Chapter 4

4. Design and Architecture

This chapter describes the overall design and architecture of the Buy Best system. It includes diagrams and explanations for system components, data structures, process flows, and design models, providing a blueprint for development and implementation.

4.1 System Architecture

The system architecture defines how the project's platform components interact with each other and with external systems. It is designed as a web-based, **multi-tier architecture** consisting of the following layers.

4.1.1 Architecture layers

The **BuyBest** system follows a layered architecture design to ensure scalability, maintainability, and efficient data processing. Each layer is responsible for specific tasks and communicates with other layers to deliver complete system functionality. This separation of concerns helps in easier development, testing, and future enhancements.

Presentation Layer (Frontend): The presentation layer handles user interaction through a responsive web interface. It allows users to search, filter, compare, and view products with ease. The interface is designed for both desktop and mobile devices. It focuses on user-friendly navigation and clear display of information.

Application Layer (Backend): The application layer processes user requests and implements system logic. It handles search operations, filtering, comparison, and user authentication. This layer also manages AI-based recommendations. It ensures smooth communication between the front end and database.

Data Layer (Database/Storage): The data layer stores all system data including products and user information. It maintains product details such as price, specifications, and ratings. It also stores user accounts, bookmarks, and alerts. This layer ensures fast and secure data access.

Integration Layer (APIs/Web Scrapers): The integration layer connects the system with external e-commerce platforms as shown in Figure . It collects product data using **APIs** and **web scraping** techniques. This layer ensures that product information is updated regularly. It also standardizes data from different sources for consistency.

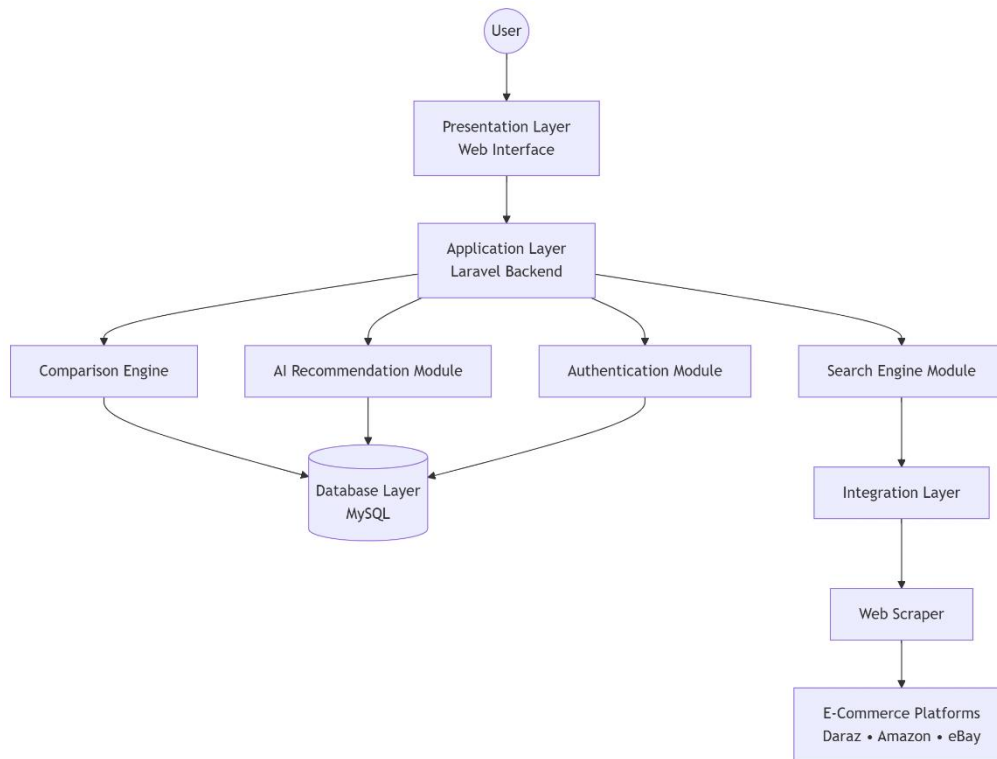


Figure 4.1 Design and Architecture Diagram

4.1.2 User-Side Architecture

The user-side architecture represents the frontend of the **Buy Best** system, which is accessed through a web browser. It provides a responsive and interactive interface for users on desktop and laptop devices. The design focuses on simplicity, usability, and smooth navigation.

- **User Interface Components:** This layer includes components such as the search bar, product listings, filters, comparison view, and login forms. Users can easily interact with these elements to perform different actions. The interface ensures clear presentation of product information.

- **User Interaction Handling:** The frontend captures user inputs such as search queries, filter selections, and comparison requests. These inputs are sent to the server for processing. The system ensures quick response and smooth interaction.
- **Display of Results:** The user-side displays results received from the backend, including product details, comparison tables, recommendations, and alerts. It also supports bookmarking and personalized features. This improves the overall user experience.

4.1.3 Server-Side Architecture

The server-side architecture represents the backend of the **Buy Best** system. It is responsible for processing user requests and managing system operations. This layer ensures proper functioning of all core features.

- **Business Logic Processing:** The backend handles system logic such as product search, filtering, comparison, and authentication. It processes user requests and generates appropriate responses. This ensures accurate and efficient system behavior.
- **Data Management:** The server communicates with the database to store and retrieve data. It manages product information, user accounts, bookmarks, and alerts. This ensures data consistency and availability.
- **Integration and AI Processing:** The server connects with external platforms using APIs and web scraping to collect product data. It also runs the AI-based recommendation module to provide personalized suggestions. This enhances system intelligence and performance.

4.2 Data Representation

The data model ensures organized storage of product and user information, enabling efficient retrieval and AI-based recommendation generation. Relationships between users, products, and alerts allow the system to provide personalized experiences and real-time notifications.

The data model ensures organized storage and efficient management of product and user information within the Buy Best system as illustrated in Figure 4.2. It provides a structured way to handle large amounts of data collected from multiple e-commerce platforms. The model maintains clear relationships between users, products, and alerts to support system functionality. This structure helps in fast data retrieval and smooth processing of user requests.

It also plays a key role in improving the performance of search, comparison, and recommendation features. The organized data supports AI-based personalized recommendations based on user behavior. It also ensures data consistency, accuracy, and scalability across the system. Overall, it enhances system reliability and improves user experience significantly.

- Stores user profiles including login details, preferences, and activity history
- Maintains product data such as price, specifications, ratings, and availability
- Manages bookmarked products for quick future access
- Stores and tracks price alerts set by users
- Supports AI recommendation engine using structured datasets
- Ensures real-time synchronization with external e-commerce APIs
- Maintains data consistency across all system modules
- Improves search efficiency by organizing product information

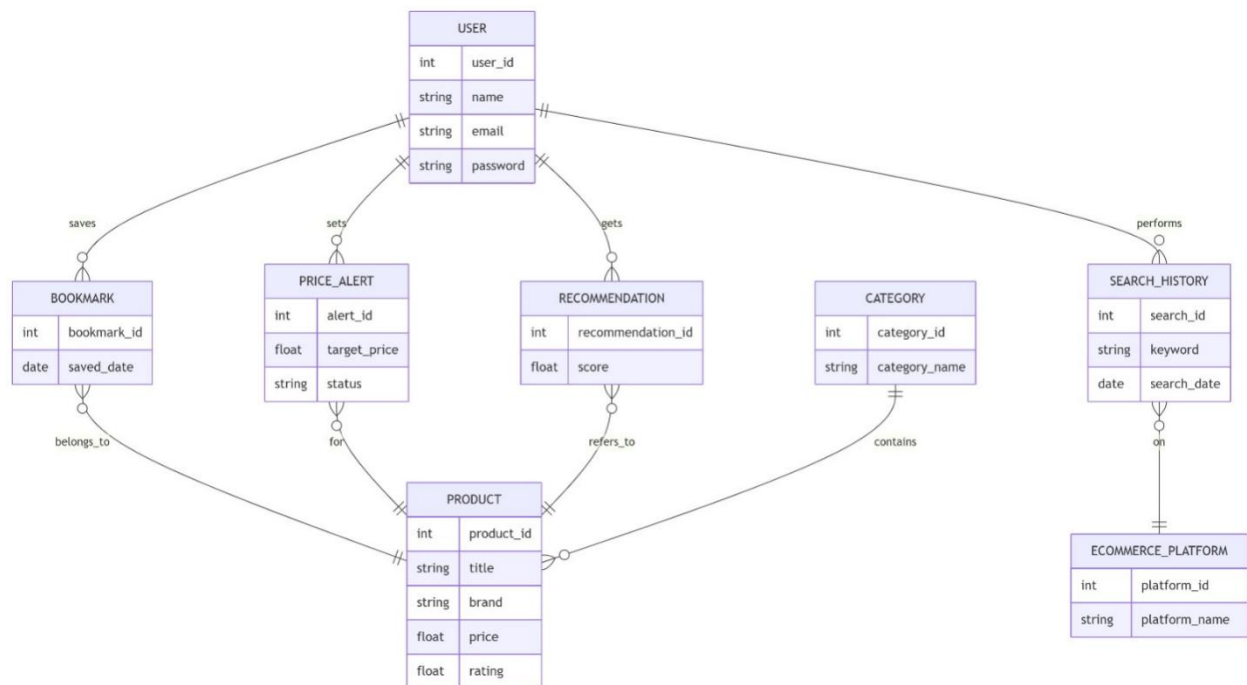


Figure 4.2 Data Representation

Table 4.1 User

Field	Description	Data Type
User ID	Unique identifier for each user.	Int
Name	Name of the user.	String
Email	Email address used for login and communication.	String
Preferences	Stores user preferences such as brands or categories.	String
Bookmarks	List of saved or bookmarked products.	List

Table 4.2 Product

Field	Description	Data Type
Product ID	Unique identifier for each product	Int
Name	Name of the product	String
Price	Current price of the product	Float
Specification	Technical details of the product	String
Ratings	Customer rating of the product	Float
Source Platform	Name of the e-commerce platform	String

Table 4.3 Comparison

Field	Description	Data Type
Comparison ID	Unique identifier for each comparison	Int
Product IDs	List of product IDs selected for comparison	List
Metrics	Criteria used for comparison such as price or features	String

Table 4.4 Alert

Field	Description	Data Type
Alert ID	Unique identifier for each alert	Int
User ID	ID of the user who set the alert	Int
Product ID	ID of the product being monitored	Int
Price Threshold	Price limit set by the user	Float
Status	Status of the alert (active/inactive)	String

4.3 Process Flow / Representation

The Process Flow / Representation describes how the Buy Best system processes a user's search request step by step, from input to final output as shown in Figure . It explains how data is collected from multiple sources, refined, and presented to the user in a meaningful way. The flow also highlights how AI is used to improve recommendations based on user behavior. Overall, it shows the complete working sequence of the system in a structured manner. It also ensures that users get accurate and updated product information in real time. The process improves efficiency by reducing the need to visit multiple platforms separately. It further enhances decision-making by providing filtered, compared, and personalized results.

4.3.1 User Query Input

- User enters search query in the system
- Query may include product name, brand, or features
- System captures input through search interface
- Input acts as trigger for processing flow
- Query is sent to backend system

4.3.2 Query Validation and Data Fetching

- System checks query for correctness and relevance
- Invalid queries are rejected by system
- Valid query is processed for data retrieval
- Product data is fetched from multiple platforms
- APIs and web scraping are used for data collection

4.3.3 Data Aggregation and Filtering

- Data from different sources is combined into one format
- Duplicate and irrelevant data is removed
- Filters are applied based on user preferences
- Data is sorted by price, brand, and specifications
- Clean dataset is prepared for display

4.3.4 AI-Based Recommendation Processing

- AI engine analyzes user behavior and history
- User preferences are identified from activity data
- Similar and alternative products are suggested
- Results improve decision-making process

4.3.5 User Interaction and Actions

- Final results are shown to user interface
- User views product details and specifications
- Products can be compared side-by-side
- Users can bookmark or save products

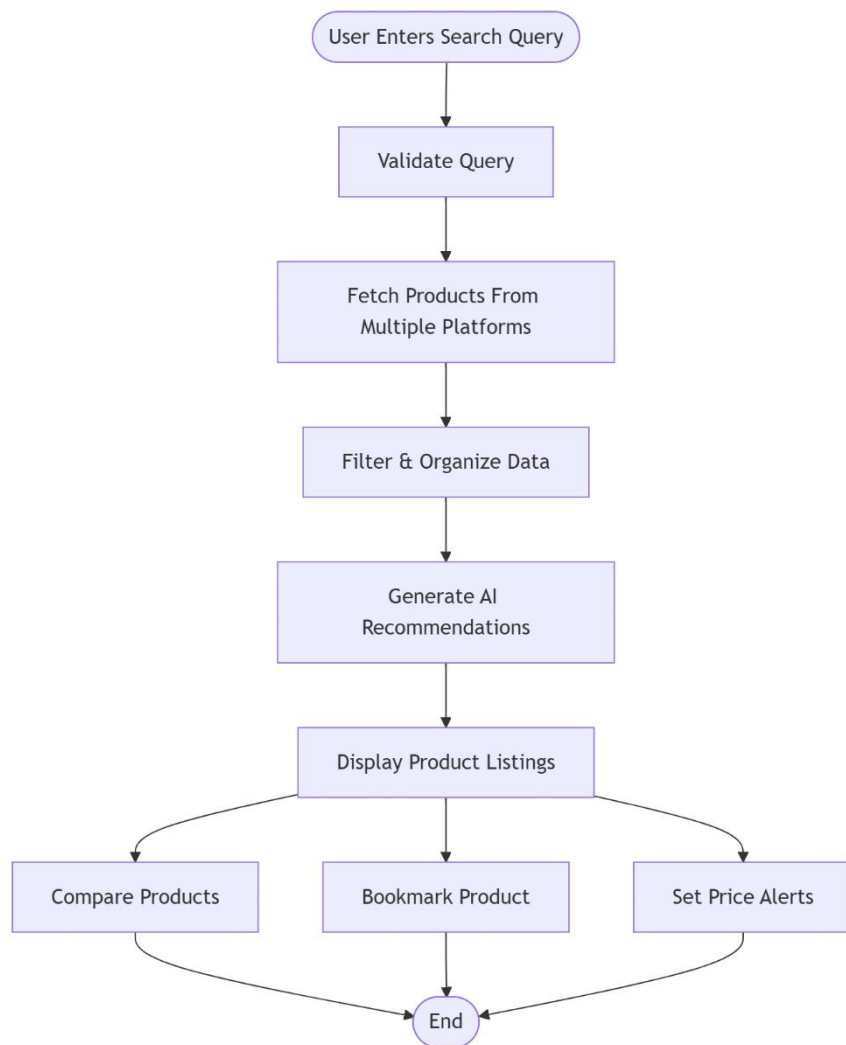


Figure 4.3 Process flow diagram

4.3.6 Data Dictionary (Buy Best Process Flow)

The Data Dictionary for the Buy Best Process Flow provides a structured description of all key processes involved in the system workflow. It defines each step from user input to final output, along with its input, output, and responsible actor. This helps in understanding how data moves through the system and how different modules interact with each other. It also ensures clarity for system design, development, and documentation purposes.

4.4 Design Models

Design models help developers understand the internal structure, behavior, and interactions of the system components as illustrated in Figure They ensure that the Buy Best system is properly aligned with the functional requirements and architecture design. These models also support system development, testing, and future enhancement by clearly defining how different modules communicate and work together.

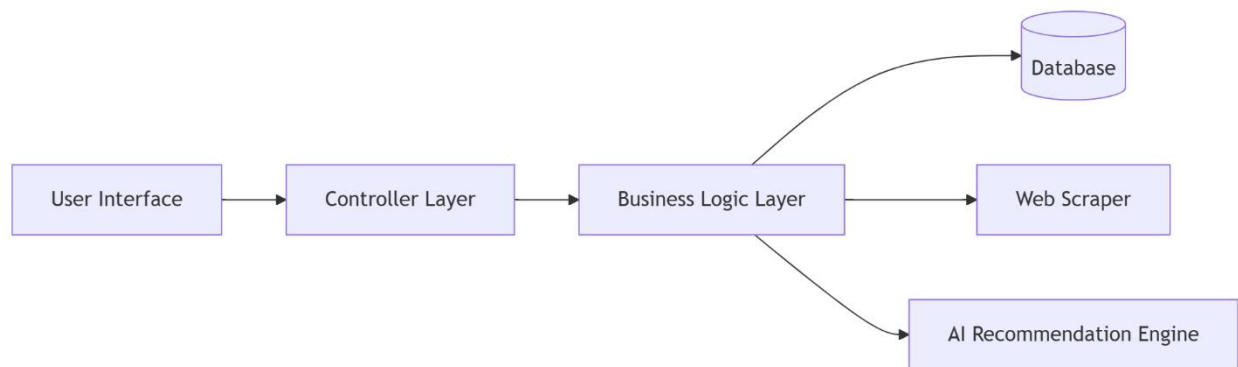


Figure 4.4 Design Model Diagram

4.1.1 Sequence Diagram

The Sequence Diagram illustrates how different components of the Buy Best system interact with each other in a time-ordered sequence as shown in Figure . This diagram helps in understanding the dynamic behavior of the system during runtime. It clearly defines communication between the user interface, backend services, and external data sources.

4.1.1.1 User Search Sequence

- User enters a product name or keyword in the search bar

- Search request is sent to the search engine module
- System retrieves product data from multiple e-commerce sources
- Filtered results are returned to the user interface

4.1.1.2 Product Comparison Sequence

- User selects multiple products for comparison
- Request is sent to the comparison engine
- System fetches detailed product specifications
- Comparison table is generated automatically

4.1.1.3 Recommendation Sequence

- System monitors user browsing and search behavior
- Recommendation engine analyzes collected user data
- AI generates personalized product suggestions
- Recommendations are updated dynamically

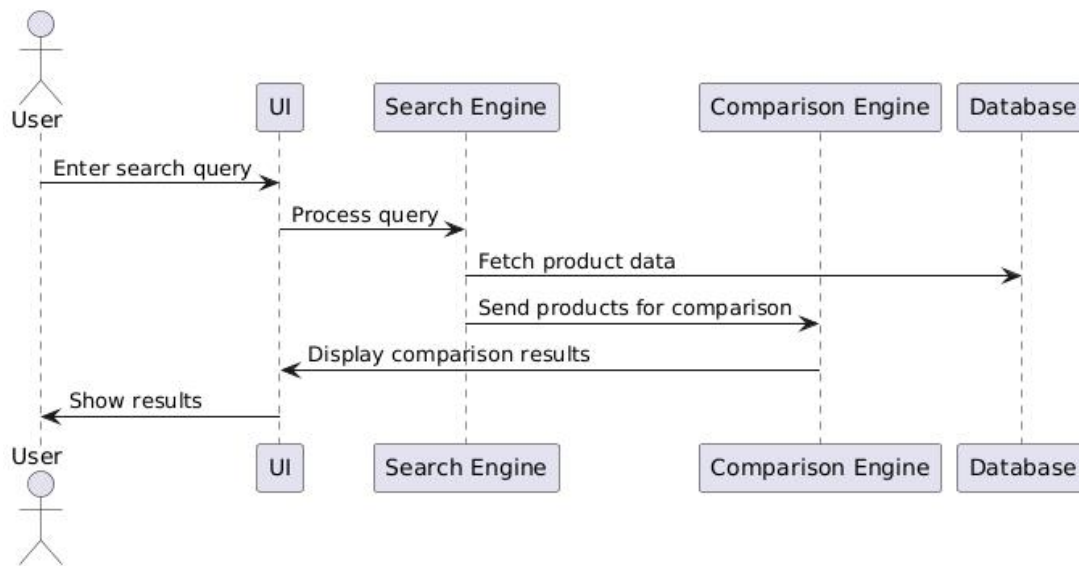


Figure 4.5 Sequence Diagram

4.1.2 Class Diagram

This Figure 4.6 represents the static structure of our project by showing classes, attributes, and relationships between objects. It defines how data is organized within the system and

how different entities interact with each other. This diagram helps in understanding system design at the object-oriented level. It is essential for database design and system implementation planning.

4.1.2.1 User Class

- Stores user ID, name, email, and preferences
- Handles login and authentication functions
- Maintains bookmarked products list
- Stores search history and activity data
- Interacts with search and recommendation modules

4.1.2.2 Product Class

- Contains product ID, name, price, and specifications
- Stores ratings and reviews from platforms
- Links product to multiple e-commerce sources
- Used in search and comparison operations
- Provides structured product information

4.1.2.3 Admin Class

- Manages users and system configurations
- Controls product data sources and updates
- Monitors system performance and analytics
- Generate reports for system evaluation
- Ensures system security and integrity

4.1.2.4 Alert Class

- Stores alert ID and user reference
- Tracks product price changes
- Maintains price threshold set by user
- Sends notifications when conditions are met

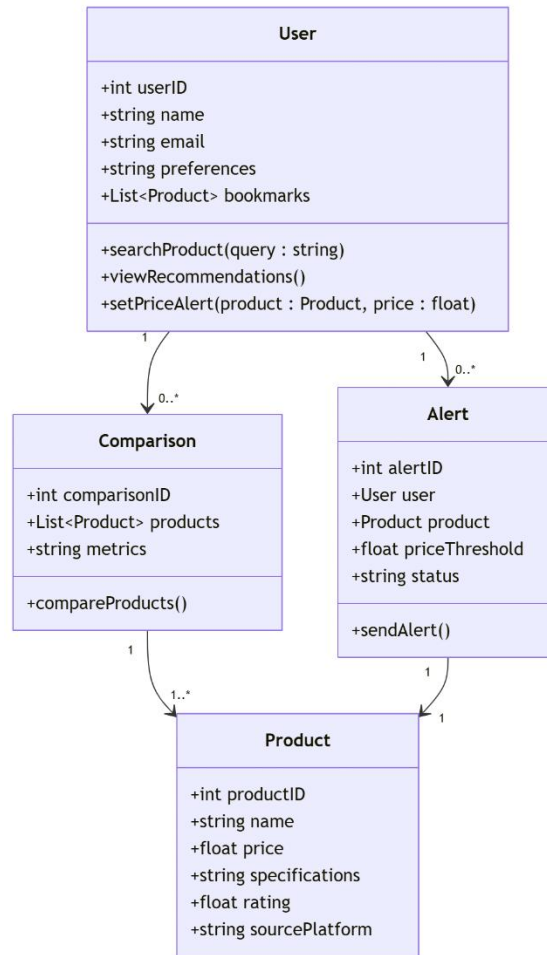


Figure 4.6 Class Diagram

4.1.3 Data flow diagram

The Figure shows how data moves through different processes of the **BuyBest** system. It represents the flow of information between users, processes, data stores, and external systems. This diagram helps in understanding how input data is transformed into meaningful output. It highlights system input, processing stages, and final results. It is useful for analyzing system efficiency and data management.

4.1.3.1 User Input Process

- User enters search query in system interface
- Input is validated by the system
- Query is forwarded to processing module
- System identifies relevant data sources
- Process initiates product retrieval

4.1.3.2 Data Processing Flow

- System collects product data from multiple platforms
- Data is cleaned and standardized
- Filtering is applied based on user preferences
- AI processes data for recommendations
- Final structured data is prepared

4.1.3.3 Data Storage Flow

- Product data is stored in database
- User profiles and preferences are saved
- Bookmarked items are recorded
- Alert settings are stored securely
- System logs are maintained

4.1.3.4 Output Generation Flow

- Processed results are sent to UI
- Product listings are displayed to user
- Comparison tables are generated
- Recommendations are shown dynamically

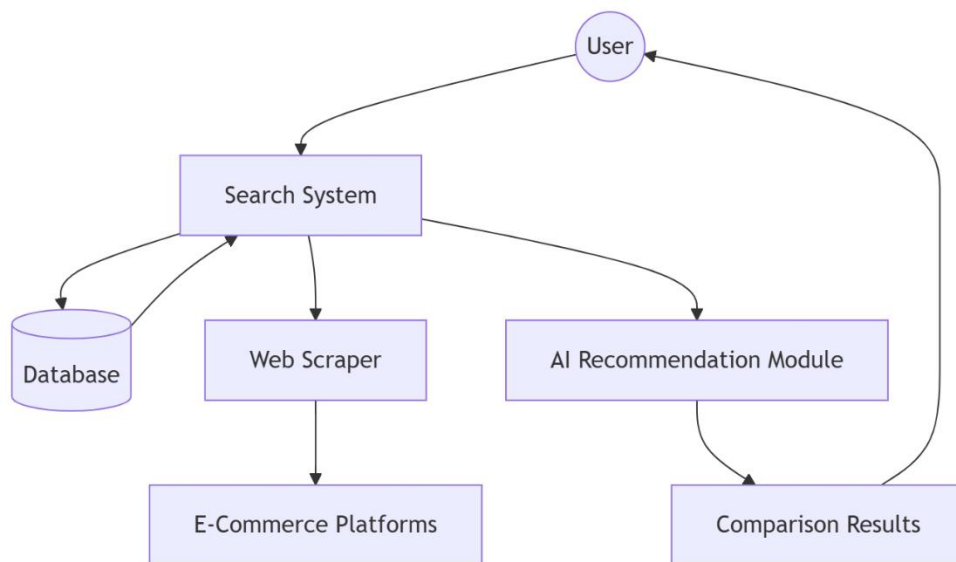


Figure 4.7 Data Flow Diagram

4.1.4 State transition diagram

The Figure 4.8 State Transition Diagram represents the different states of the **BuyBest** system objects and how they change from one state to another. It shows how the system responds to user actions and system events over time. This diagram is useful for understanding object behavior and lifecycle management. It clearly defines transitions between different states such as idle, processing, and completed. It helps in designing interactive and responsive system behavior.

4.1.4.1 Search State Flow

- System is in idle state waiting for user input
- User enters search query triggering processing state
- System fetches and filters product data
- Results are generated and displayed
- System returns to idle state

4.1.4.2 Product View State Flow

- User selects a product from listings
- System enters product detail state, detailed information is displayed
- User may compare or bookmark product
- System returns to browsing state

4.1.4.3 Comparison State Flow

- User selects multiple products
- Data is retrieved and analyzed
- Comparison results are displayed
- System returns to selection state

4.1.4.4 Alert State Flow

- User sets price alert for product
- System enters monitoring state
- Continuous price tracking occurs

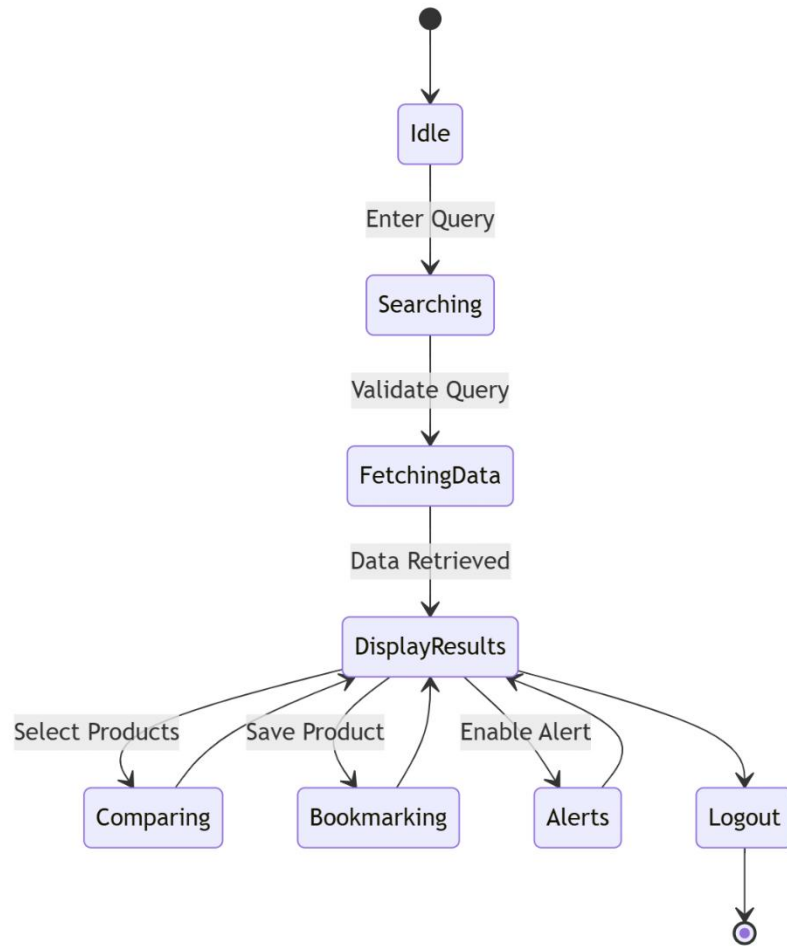


Figure 4.8 State Transition Diagram

Chapter 5

5. Implementation

This chapter describes the implementation approach for the **BuyBest** system, including algorithms, external APIs, and user interface design. It explains how different system components are developed and integrated to perform required functionalities. The focus is on implementing core modules such as product search, comparison, recommendations, and alerts. It also highlights how frontend and backend systems communicate efficiently. The chapter covers techniques used to ensure system reliability and performance. It discusses handling of errors and system scalability for future growth. Overall, it provides a clear view of how the system is practically built and executed.

5.1 Implementation Strategy

The implementation strategy defines how the **BuyBest** system is developed and deployed effectively. It focuses on dividing the system into frontend and backend components for better organization. The strategy ensures smooth interaction between user interface and server-side logic. It also considers performance optimization and efficient data handling. Proper techniques are used to handle errors and unexpected system behavior. Scalability is considered to support increasing users and data in future. Overall, the strategy ensures a reliable, maintainable, and high-performing system.

5.1.1 Frontend

- Developed using responsive web technologies for user interaction
- Provides interface for search, filter, comparison, and alerts
- Ensures smooth navigation and user-friendly design
- Displays product data, recommendations, and notifications
- Handles user inputs and sends requests to backend

5.1.2 Backend

- Processes user requests and implements business logic
- Handles search, filtering, comparison, and recommendations
- Communicates with database for storing and retrieving data
- Integrates APIs and web scraping for product data collection

- Manages user authentication and session control

5.1.3 Error Handling

- Validates user input to prevent incorrect data processing
- Displays error messages for invalid or empty queries
- Handles API failures and data fetching errors
- Ensures system stability during unexpected conditions
- Logs errors for debugging and system improvement

5.1.4 Scalability and Maintenance

- Designed to handle increasing number of users and data
- Supports addition of new features and modules easily
- Uses modular architecture for easier maintenance
- Ensures efficient database management and performance
- Allows regular updates without affecting system stability

5.2 Algorithm

The algorithm defines the step-by-step process used by the **BuyBest** system to search, process, and present product data. It ensures efficient handling of user queries and accurate retrieval of product information from multiple platforms. The algorithm integrates data processing, filtering, and comparison in a structured flow. It also incorporates AI techniques to generate personalized recommendations. Overall, it improves system performance, accuracy, and user experience by automating complex operations.

5.1.1 Purpose

The purpose of this algorithm is to fetch product data from multiple e-commerce platforms, process and organize it into a structured format, and present it to users in a meaningful way. It enables users to search for products using keywords, brands, or specifications and receive accurate results from different sources. The algorithm ensures that data collected from various platforms is cleaned, standardized, and filtered based on user preferences. It also supports product comparison by analyzing multiple product attributes side by side. Additionally, it integrates an AI-based recommendation system that suggests relevant and

alternative products based on user behavior. The algorithm is designed to improve efficiency, reduce search time, and enhance decision-making. Overall, it provides a complete workflow from user input to final output in an optimized and user-friendly manner.

5.1.2 Input

- User search query including product name, brand, or specifications
- User preferences such as price range, ratings, and features
- Data retrieved from external e-commerce platforms
- User interaction history and behavior data
- Filter parameters selected by the user
- System configuration and API responses

5.1.3 Output

- Aggregated list of products from multiple platforms
- Filtered and sorted product results
- Detailed product information including specifications and pricing
- Side-by-side comparison results of selected products
- AI-based personalized recommendations
- Notifications or alerts based on user preferences

5.1.4 Algorithm Steps

The process begins when the user enters a search query into the system as shown in Figure .The system validates the input and sends requests to multiple e-commerce platforms to collect product data. The retrieved data is then cleaned, organized, and filtered according to user preferences. After processing, relevant results are displayed, and users can perform actions such as comparison or bookmarking. The system also analyzes user behavior to generate personalized recommendations.

- User enters a search query into the system
- System validates the query for accuracy and completeness
- Product data is fetched from multiple platforms
- Data is cleaned and duplicates are removed
- Product information is organized into a structured format

- Filters are applied based on user preferences
- Relevant products are sorted and displayed
- User selects products for comparison
- System generates comparison results
- AI engine analyzes behavior and generates recommendations

5.1.5 AI Recommendation Logic

- Collect user search history and interaction data
- Identify user preferences such as brand, category, and price range
- Analyze product features and similarities
- Rank products based on relevance and user interest
- Suggest alternative and trending products
- Continuously update recommendations using new data

5.1.6 Data Processing and Optimization

- Process data in real-time for quick response
- Use efficient filtering techniques to reduce data load
- Apply caching for frequently searched products
- Optimize API calls to reduce response time
- Store structured data for faster retrieval
- Ensure scalability for handling large datasets

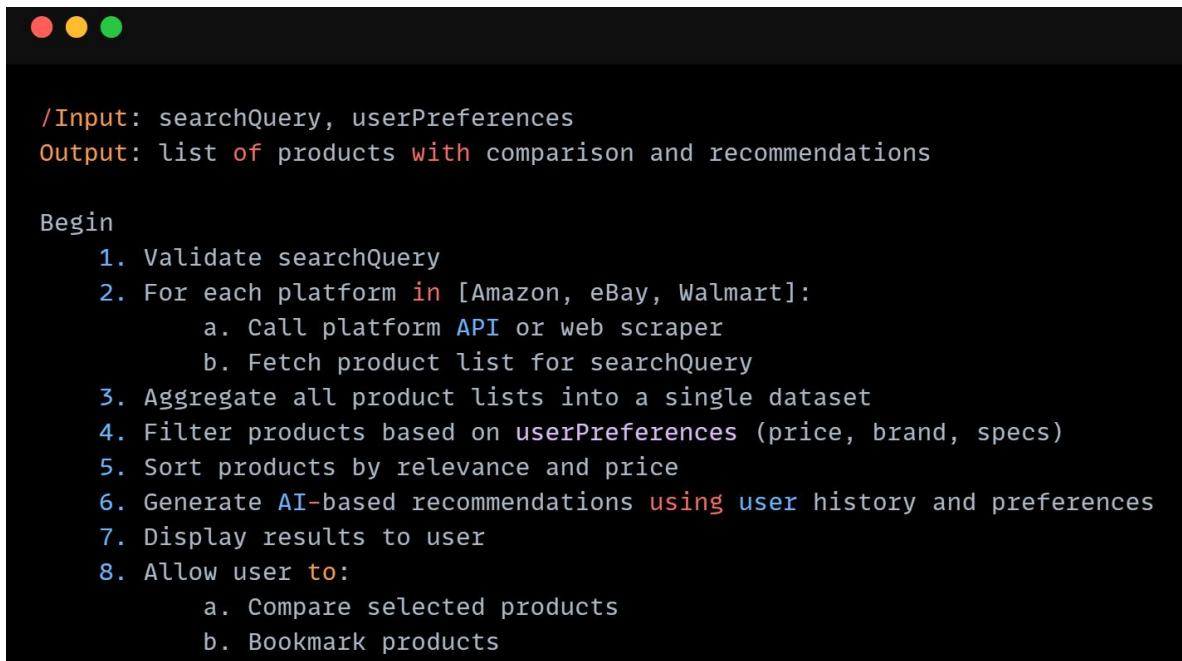
5.1.7 Error Handling in Algorithm

- Validate input to prevent incorrect queries
- Handle empty or null search results
- Manage API failures and timeouts
- Display meaningful error messages to users
- Ensure system continues running during failures
- Log errors for debugging and improvement

5.1.8 Result Generation

- Display product list in structured format

- Provide clear comparison results
- Highlight recommended products
- Allow user interaction with results
- Enable bookmarking and alert features
- Ensure fast and responsive output display



```

/Input: searchQuery, userPreferences
Output: list of products with comparison and recommendations

Begin
  1. Validate searchQuery
  2. For each platform in [Amazon, eBay, Walmart]:
      a. Call platform API or web scraper
      b. Fetch product list for searchQuery
  3. Aggregate all product lists into a single dataset
  4. Filter products based on userPreferences (price, brand, specs)
  5. Sort products by relevance and price
  6. Generate AI-based recommendations using user history and preferences
  7. Display results to user
  8. Allow user to:
      a. Compare selected products
      b. Bookmark products

```

Figure 5.1 Search Query Object

5.2 External APIs

External APIs are used to fetch product data from multiple e-commerce platforms and provide additional functionality such as notifications and recommendations. They allow the system to collect real-time product information and ensure that users receive updated and accurate data. These APIs help in integrating different platforms into a single system, enabling easy comparison of products across websites. They also support advanced features like AI-based recommendations and alert notifications. Overall, external APIs play a key role in improving system efficiency, automation, and user experience.

5.2.1 Data Fetching APIs

- Used to retrieve product details from multiple e-commerce platforms

- Provide data such as price, specifications, ratings, and availability
- Ensure real-time updates of product information
- Support integration with different online stores
- Help in aggregating data into a unified format

5.2.2 Recommendation APIs

- Used to generate AI-based personalized suggestions
- Analyze user behavior and search history
- Suggest relevant and alternative products
- Improve user experience and engagement
- Continuously update recommendations based on new data

5.2.3 Notification APIs

- Used to send alerts and updates to users
- Notify users about price changes and product availability
- Support real-time communication with users
- Enable cross-platform alert delivery
- Improve responsiveness of the system

5.2.4 Integration and Processing

- APIs connect the system with external platforms
- Data is processed and standardized after retrieval
- Ensures compatibility between different data sources
- Reduces manual effort in data collection
- Supports smooth communication between modules

5.2.5 Benefits of Using APIs

- Provides real-time and accurate product information
- Enables cross-platform product comparison
- Improves system scalability and flexibility
- Reduces development time by using external services
- Enhances overall system performance and reliability

Table 5.1 Data dictionary

Name of API	Description of API	Purpose of Usage	Function/Class Name
Amazon product API	Provide product details from Amazon	Fetch product Name, Price, Specs, Ratings	AmazonScraper.fetchProducts()
eBay API	Provides product listings from eBay	Fetch product Data for comparison	eBayScraper.fetchProducts()
Walmart API	Provides Walmart product data	Retrieve product information for aggregation	WalmartScraper.fetchProducts()
Notification API	Send email or notify	Alert users about price changes	AlertManager.sendAlert()
Recommendation API	AI module for product suggestions	Generate personalized recommendations	RecommendationEngine.generate()

5.3 User Interface

The user interface of the **BuyBest** system is designed to provide a smooth, intuitive, and efficient user experience by focusing on simplicity and clarity. It minimizes user effort by reducing unnecessary clicks and presenting information in a well-structured and organized layout. The interface follows a responsive design approach, ensuring compatibility across

desktops, laptops, and mobile devices. It emphasizes easy navigation through clearly defined menus, buttons, and sections, allowing users to quickly access different features of the system. Visual elements such as icons, cards, and color highlights are used to improve readability and guide user actions effectively. The UI also supports interactive features like filtering, comparison, and bookmarking to enhance usability. Consistency in design across all screens ensures a seamless user experience. Overall, the interface improves efficiency, user satisfaction, and accessibility while helping users perform tasks like searching, comparing, and managing products with ease.

5.3.1 Search Screen

The search screen is the main entry point of the system where users begin their interaction as shown in Figure 5.2. It features a modern and visually appealing design with a focus on simplicity and usability. The layout highlights the search functionality prominently to ensure quick access.

Navigation elements are clearly placed to guide users across different sections of the system.

- Gradient hero section with centered search bar
- Navigation bar with links to key features
- Login button for user authentication
- Feature cards for AI recommendations and comparison

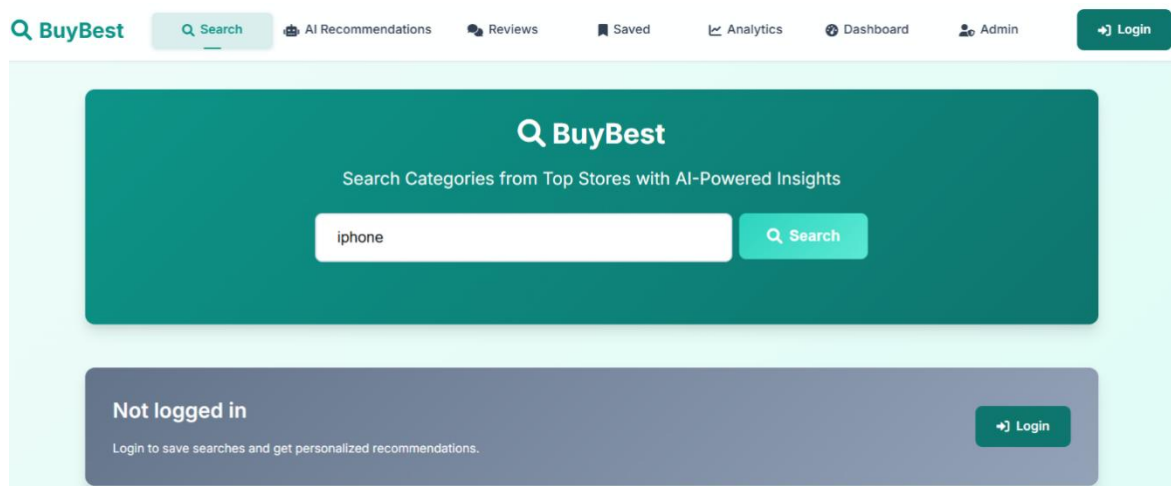


Figure 5.2 Search Screen

5.3.2 Login Screen

The login screen provides a secure and user-friendly authentication system for both new and existing users as shown in Figure 5.3. It is designed with a clean two-column layout to separate login and registration functionalities. The interface ensures that users can easily access their accounts or create new ones.

It maintains simplicity while ensuring proper data input and validation.

- Two-column layout for login and registration
- Input fields for email, password, and user details
- “Remember Me” option for convenience
- Clear separation between login and signup sections
- Simple and organized interface design

The screenshot displays the 'BuyBest' login and registration interface. At the top, a navigation bar features the 'BuyBest' logo, a search icon, and links for 'AI Recommendations', 'Reviews', 'Saved', 'Analytics', 'Dashboard', and 'Admin'. A 'Login' button is positioned on the right. The main content area is divided into two columns. The left column, titled 'Login', contains an 'Email Address' field with 'admin@example.com', a 'Password' field with masked characters, a 'Remember me' checkbox, and a 'Login' button. The right column, titled 'Create Account', includes a 'Full Name' field with the placeholder 'Enter your full name', an 'Email Address' field with 'admin@example.com', a 'Password' field with masked characters, a 'Confirm Password' field with the placeholder 'Confirm your password', a checkbox for agreeing to the 'Terms of Service' and 'Privacy Policy', and a 'Create Account' button.

Figure 5.3 Login Screen

5.3.3 Search Results Screen

The search results screen displays products in a structured and easy-to-browse format as shown in Figure 5.4. It allows users to quickly scan multiple products and compare their key attributes. The design focuses on clarity and accessibility of product information.

It also supports quick actions for better user interaction.

- Grid layout for displaying multiple products
- Product cards with image, name, brand, and rating
- Action buttons like “Compare Selected” and “Save All”
- Spacious layout for easy browsing
- Consistent header for navigation

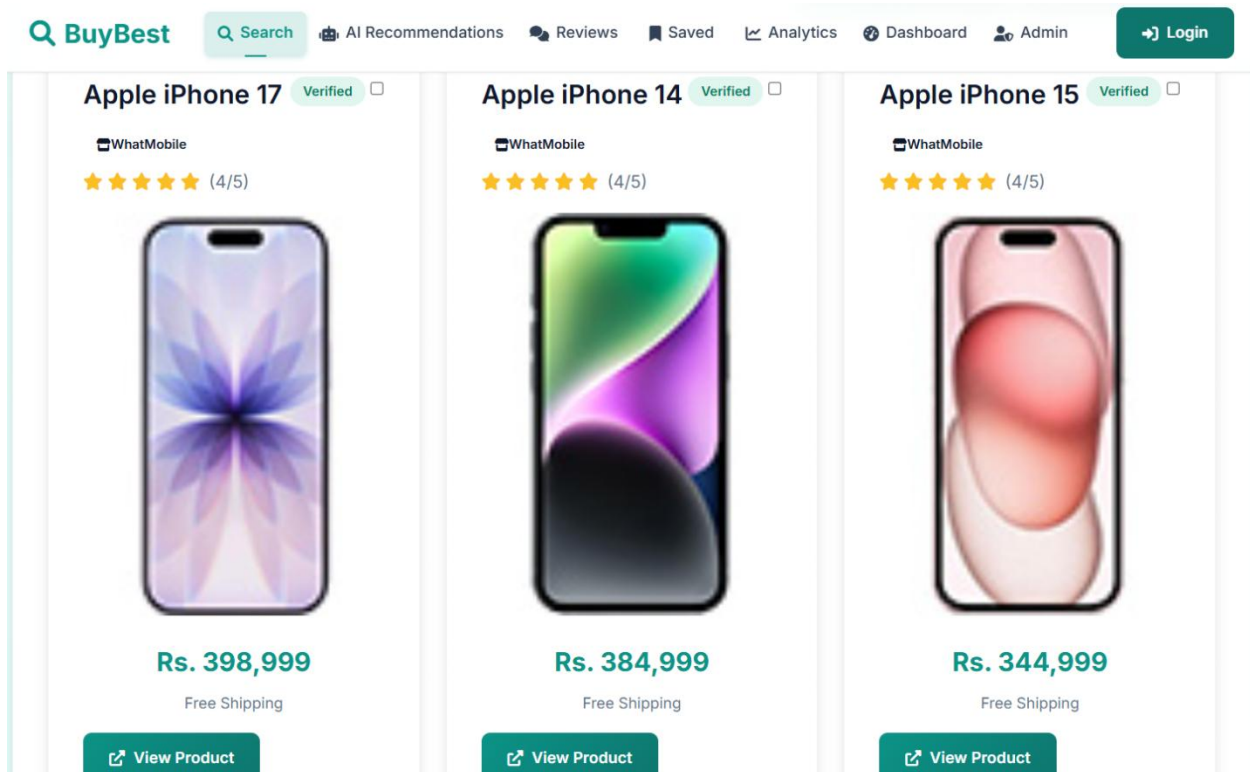


Figure 5.4 Search Result Screen

5.3.4 Product Details Screen

The product details screen provides complete information about a selected product as shown in Figure 5.5. It redirects users to the original e-commerce platform for authenticity. This ensures that users receive accurate and up-to-date product details directly from the source.

The layout follows a standard e-commerce design for familiarity.

- Displays product images, specifications, and pricing
- Redirects to original seller website
- Shows detailed descriptions and features
- Includes categories and related information
- Ensures authentic product data



Figure 5.5 Product Details Screen

5.3.5 Product Comparison Screen

The product comparison screen helps users analyze multiple products side by side as shown in Figure 5.6. It presents key product attributes in a structured table format. This makes it easier for users to identify differences and similarities.

The design is kept simple for better readability and understanding.

- Table layout for side-by-side comparison

- Displays price, rating, and specifications
- Option to add or remove products
- Export comparison data feature
- Clean and minimal design

Features	Apple iPhone 17	Apple iPhone 14
	Rs. 398,999 ★★★★☆4.0	Rs. 384,999 ★★★★☆4.0
Display	6.1-6.7 inch Super Retina XDR	6.1-6.7 inch Super Retina XDR
Processor	Apple A17 Pro	Apple A15/A16 Bionic
RAM	6GB-8GB	6GB
Storage	128GB-1TB	128GB-1TB
Camera	48MP Main + 12MP Ultra Wide	12MP-48MP Main + 12MP Ultra Wide
Battery	Up to 26-29 hours	Up to 20-29 hours
OS	iOS 17	iOS 15-17
Price	Rs. 398,999	Rs. 384,999

Figure 5.6 Product Comparison Screen

5.3.6 AI Recommendations Screen

This screen provides intelligent product suggestions based on user behavior and preferences as shown in Figure 5.7. It uses AI techniques to analyze data and generate meaningful recommendations. The layout highlights important product metrics for better decision-making.

Visual elements enhance the understanding of recommendations.

- Displays personalized product suggestions
- Highlights price, rating, and performance metrics
- Uses visual indicators and banners
- Simple and focused layout design

- Helps users choose better alternatives

The screenshot displays the 'BuyBest' application interface. At the top, a navigation bar includes a search icon, 'Search', 'AI Recommendations' (highlighted), 'Reviews', 'Saved', 'Analytics', 'Dashboard', 'Admin', and a 'Login' button. The main section is titled 'Analyze Product' with a subtext: 'Enter a product name to get AI-powered analysis including score, price estimate, rating, and recommendations.' Below this is a 'Product Name:' label and a text input field containing 'iphone 15 Pro'. A green 'Analyze Product' button is positioned below the input field. Underneath, 'Quick examples:' are listed in a grid: Samsung, Apple, Vivo, Huawei, Dell, HP, Oppo, and Xiaomi. An 'Auto-analyze:' toggle switch is turned on. A green message 'Analysis complete!' with a pencil icon is visible. The bottom section is titled 'AI Analysis Results' with a subtext: 'Comprehensive breakdown generated by our AI engine.' and the product name 'iphone 15 Pro' in green.

Figure 5.7 AI Recommendation Screen

5.3.7 Saved Searches Screen

The saved searches screen allows users to access their previously saved queries easily as shown in Figure 5.8. It helps users reuse past searches without entering data again. The layout is organized to provide quick access and management options.

This feature improves convenience and efficiency.

- Grid layout of saved search cards
- Displays keywords and saved time
- Options to search again or view results
- Export and clear all features
- Easy access to past searches

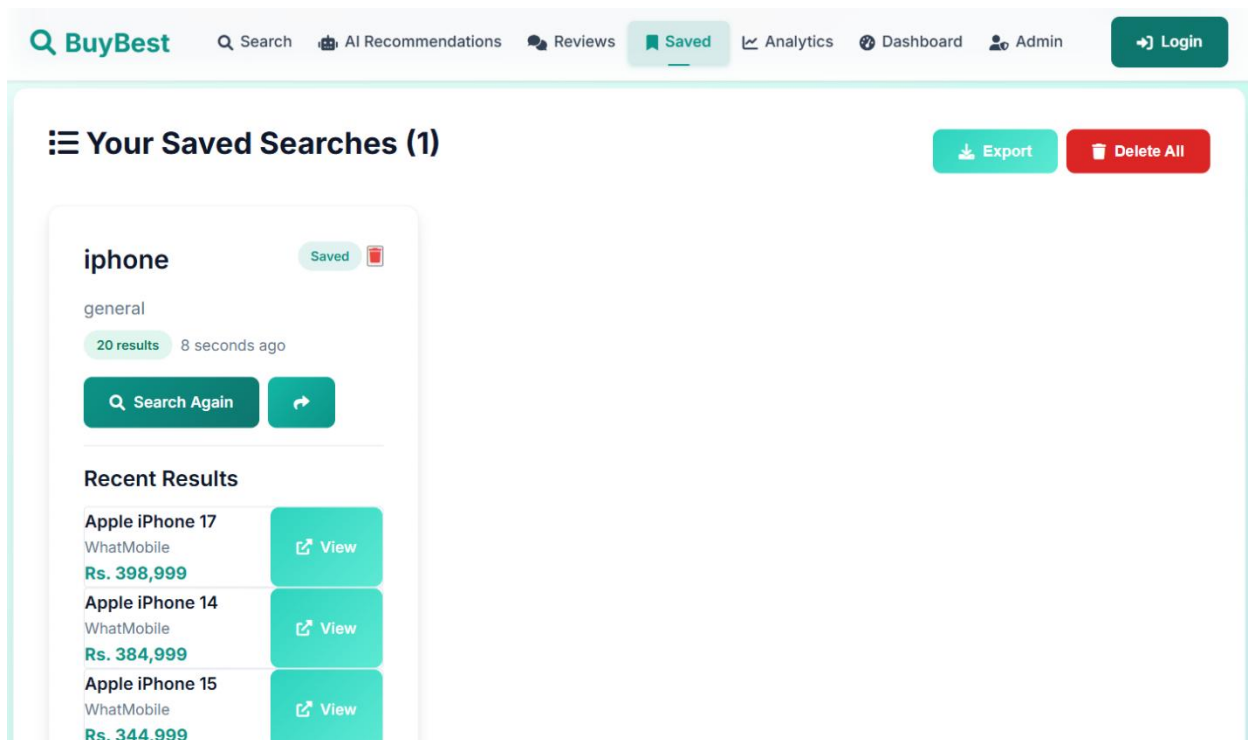


Figure 5.8 Saved Search Screen

5.3.8 Admin Panel

This Figure 5.9 shows an admin dashboard titled **BuyBest Admin** with a left sidebar menu including Dashboard, Users, Saved Searches, Messages, AI Logs, Products, and Settings. At the top of the main area, there are summary statistic cards displaying Total Searches, Registered Users, Saved Searches, Contact Messages, and AI Reviews Analyzed. Below these, two graphs visualize Searches and AI Logs with bar charts representing daily activity. At the bottom, there is a Recent Searches table listing queries, results, users, and dates, along with a link to view saved searches.

It supports decision-making through organized data presentation.

- Sidebar menu with multiple management options
- Dashboard with summary statistics cards
- Graphs for searches and AI logs

- Table for recent searches and user activity
- Access to system settings and controls

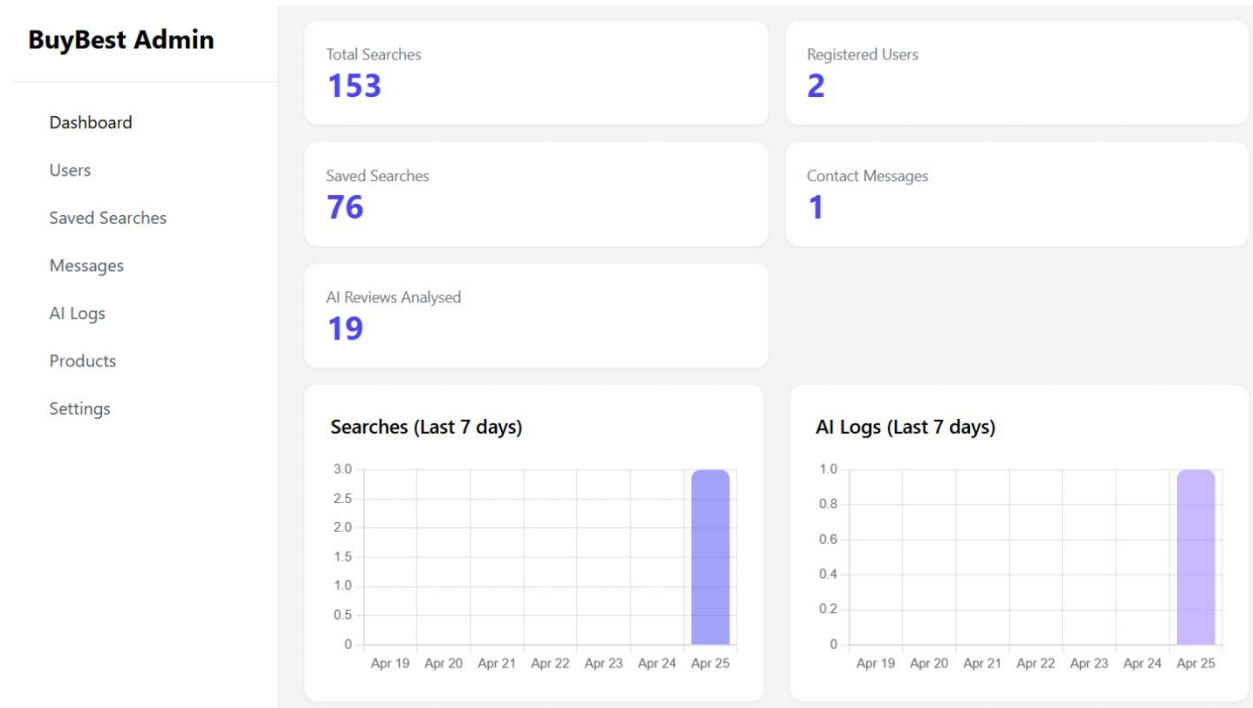


Figure 5.9 Admin Panel

5.3.9 UI Design and Consistency

The UI design of the project is developed to ensure a consistent, structured, and visually appealing experience across all modules. It focuses on maintaining uniformity in layout, colors, fonts, and interactive elements so that users can easily understand and navigate the system. Consistency in design reduces confusion and helps users quickly become familiar with system functions. The interface is built using modern UI principles that prioritize clarity, simplicity, and responsiveness. Every screen follows the same design rules to ensure a seamless transition between different sections. Standard components such as buttons, cards, and navigation bars are reused throughout the system. UI consistency improves usability [12], efficiency, and user satisfaction.

5.3.9.1 Visual Consistency

- The system uses a single-color theme across all pages to maintain a professional and unified appearance

- Fonts are standardized to ensure readability and visual harmony throughout the interface
- Icons and graphical elements remain consistent to avoid confusion between different actions
- Button styles such as shape, color, and hover effects are kept uniform across all screens
- Spacing and alignment rules are applied consistently to create a clean layout

5.3.9.2 Layout Structure

- The system uses a grid-based layout to organize product listings and information clearly
- Each page follows a structured design with defined sections for content separation
- Headers and navigation bars remain fixed in position across all pages
- Product cards and information blocks follow a consistent design pattern
- Proper use of whitespace improves readability and reduces visual clutter

5.3.9.3 Navigation Consistency

- The navigation bar remains the same on all pages for easy access to features
- Users can move between sections like search, comparison, and recommendations without confusion
- Menu items are placed in a predictable and familiar structure
- Breadcrumb navigation helps users track their location within the system
- Transitions between pages are smooth and consistent

5.3.9.4 Component Standardization

- UI components such as search bars, filters, and buttons are reused across the system
- Forms for login, registration, and input fields follow the same design pattern
- Alerts, notifications, and pop-ups have a consistent style and behavior
- Product display cards maintain a uniform structure for easy comparison
- Standardized components reduce development complexity and improve maintainability

5.3.9.5 Responsiveness

- The interface automatically adjusts to different screen sizes including mobile, tablet, and desktop
- Flexible layouts ensure that content remains readable on all devices
- Images and product cards resize properly without losing quality or structure
- Navigation adapts for mobile users using simplified menus
- The system ensures smooth user experience across all platforms

5.3.10 Interaction Design

Interaction design in the **BuyBest** system focuses on how users interact with system features and how the system responds to those actions as shown in Figure 5.10. It ensures that every user action produces clear and immediate feedback, making the system intuitive and easy to use. The design simplifies complex operations like searching, filtering, and comparing products into smooth interactions. It improves communication between the user and system through responsive UI elements. The system is designed to minimize effort while maximizing efficiency and usability. Feedback mechanisms such as alerts and notifications guide users throughout their journey. Overall, interaction design enhances engagement, usability, and system performance.

5.3.10.1 User Flow Interaction

- The system provides a step-by-step flow starting from search to product results
- Each action leads logically to the next stage without confusion
- Users can easily move between search, comparison, and recommendation screens
- The system minimizes the number of steps required to complete tasks
- Navigation flow is designed to be simple and predictable

5.3.10.2 System Feedback

- The system provides instant responses when users perform actions
- Loading indicators are displayed during data processing operations
- Success messages confirm actions like saving or bookmarking products
- Error messages guide users when incorrect input is provided
- Notifications inform users about price changes and updates

5.3.10.3 Input Interaction

- Users can enter search queries through a simple search bar
- Filters allow refinement based on price, brand, and specifications
- Dropdown menus help users select options easily
- Clicking and selecting products triggers comparison actions
- Inputs are designed to be simple and user-friendly

5.3.10.4 Output Interaction

- Product results are displayed in structured and readable formats
- Comparison tables show differences between selected products clearly
- AI recommendations are highlighted for better decision-making
- Alerts and notifications are shown in real time
- Output is designed for quick understanding and usability

5.3.10.5 Engagement Features

- Users can bookmark products for future reference
- Search history is saved for quick reuse
- Interactive comparison tools improve decision-making
- AI suggestions help users discover better options
- Price alerts keep users updated on product changes

5.3.10.6 UI Flow

The UI Flow of the **BuyBest** system represents how users interact with different screens and features in a structured sequence. It shows the flow starting from system access to searching products, viewing results, and performing actions like comparison, saving, and getting recommendations. The diagram also highlights how both users and administrators navigate through the system to perform their respective tasks efficiently.

- **System Entry and Navigation:** The flow begins when the user opens the system. The search screen is displayed as the main interface. Users can proceed as guest or logged-in user
- **Search and Processing Flow:** Users enter a query to search for products. The system fetches data from multiple platforms. Results are displayed in a structured format

- **User Actions on Results:** Users can view detailed product information. Multiple products can be selected for comparison. AI recommendations are provided based on user behavior. Users can save products or set alerts
- **Admin Access Flow:** Admin users can access the dashboard from the main flow. The dashboard provides system control and analytics. Admin can manage users and system data

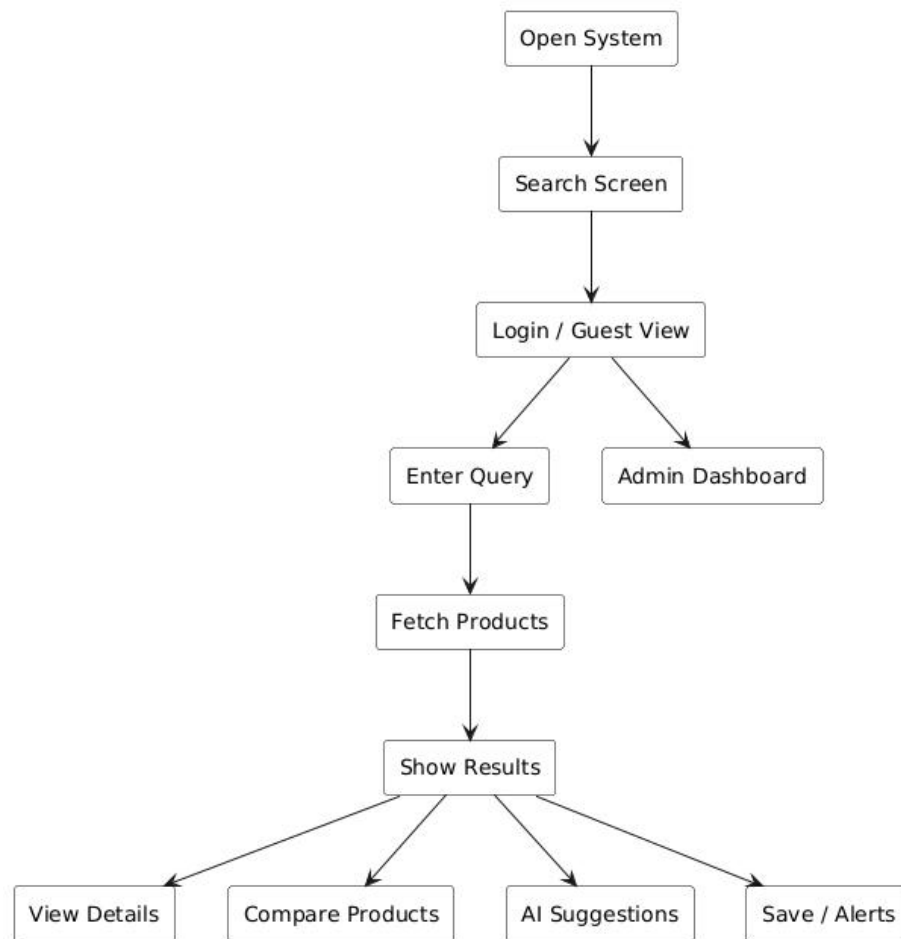


Figure 5.10 UI Flow Diagram

Chapter 6

6. Testing and Evaluation

After the successful development of the **BuyBest** system, testing was performed to ensure that the system works correctly and meets all defined requirements. Testing helps in identifying errors and improving system reliability before deployment. Different types of testing were conducted including unit testing, functional testing, and integration testing. Both manual and automated testing approaches were used to verify system performance.

6.1 Manual Testing

Manual testing is the process of testing the system by manually checking its features without using automated tools. It involves testers interacting with the system in the same way as real users to ensure that all functionalities work correctly. This type of testing is important for identifying errors, usability issues, and unexpected behavior in the system. It helps verify that features such as search, login, product comparison, and recommendations are functioning as expected. Manual testing also ensures that the system interface is user-friendly and easy to navigate. Different test cases are executed to check system behavior under various conditions and inputs. It plays a key role in validating both functional and non-functional requirements. Overall, manual testing improves system quality, reliability, and user satisfaction before final deployment.

6.1.1 System Testing

System testing was carried out to evaluate the complete system. It ensures that all modules such as search, comparison, AI recommendations, and saved searches are working together properly. This test checks whether the integrated system meets the defined requirements. It verifies that data flows correctly between different components. The system is tested under different scenarios to ensure consistent performance. Any bugs or issues identified are fixed before deployment. This ensures the system is stable and ready for real-world use.

6.1.2 Unit Testing

Unit testing is the process of testing individual components of the **BuyBest** system separately. It ensures that each feature, such as login, search, filtering, and comparipson, works correctly on its

own. Developers test each module independently to identify errors early in the development process. This testing helps in isolating issues and fixing them quickly. It improves code quality and reduces the chances of system failure later. Each unit is validated against expected outputs. Overall, unit testing ensures that every component performs its function correctly.

6.1.3 Functional Testing

Functional testing is performed to verify that all system features operate according to the specified requirements. It checks whether functions like searching, filtering, comparison, bookmarking, and alerts are working properly. Different inputs are tested to ensure correct outputs are generated. This test ensures that user actions produce expected results. It helps in identifying logical and functional errors. Overall, it confirms that the system meets its intended purpose.

6.1.4 Integration Testing

Integration testing is used to evaluate how different modules of the system work together. It ensures smooth communication between frontend, backend, database, and external APIs. This test verifies that data flows correctly across all components. It helps identify issues related to module interaction. The system is tested as a combined unit rather than individual parts. This improves system coordination and reliability.

6.1.5 User Interface Testing

User interface testing focuses on checking the design and usability of the system. It ensures that all UI elements such as buttons, forms, and menus function correctly. The layout is tested for consistency and proper alignment. It verifies that the interface is easy to use and visually clear. This testing improves overall user experience. Any design-related issues are identified and corrected.

6.1.6 Performance Testing

Performance testing evaluates how the system performs under different conditions. It checks response time, system speed, and handling of multiple users. The system is tested for delays or slow performance. This helps identify bottlenecks and optimize system efficiency. It ensures that

the system can handle real-world usage effectively. Overall, it improves system performance and stability.

6.1.7 Error Handling Testing

Error handling testing ensures that the system can manage unexpected situations properly. It checks how the system responds to invalid inputs, failed operations, and system errors. Proper error messages are displayed to guide users. The system is tested to ensure it does not crash during failures. Logs are maintained for debugging purposes. This improves system reliability and robustness.

6.1.8 Unit Testing 1: User Login

The purpose of this unit test is to ensure that the user login functionality of the **BuyBest** system is working correctly and securely. It verifies that users can log into the system using valid credentials and access their personalized features. The test checks whether the system properly validates user input such as email and password. It also ensures that incorrect credentials are handled appropriately by displaying error messages. This testing helps identify issues related to authentication at an early stage. Overall, it ensures that the login module is reliable, secure, and user-friendly.

6.1.8.1 Test Objective

The objective of this test is to verify that the login functionality of the system works correctly and efficiently. It ensures that only valid users are able to access the system using correct credentials. The test also checks the validation of input fields such as email and password to prevent incorrect data entry. Additionally, it confirms that users are properly redirected to the dashboard or main interface after a successful login.

6.1.8.2 Test Inputs

The test inputs include different types of user credentials to evaluate system behavior. These include valid email and password combinations to test successful login. Invalid email or password inputs are used to check error handling. Empty input fields are tested to ensure proper validation messages are displayed. Incorrect email formats are also used to verify input validation rules.

6.1.8.3 Expected Output

The expected output of this test is that users with valid credentials are able to log in successfully. In case of invalid login details, the system should display appropriate error messages. If the input fields are left empty, the system should show warning messages to guide the user. After successful login, the user should be redirected to the dashboard or main system interface.

6.1.8.4 Test Procedure

The test procedure begins by entering valid login credentials into the system. The login button is then clicked to submit the information. The system response is observed to verify correct behavior. The same steps are repeated using invalid credentials and empty fields to test error handling and validation. This ensures that all possible scenarios are properly tested.

6.1.8.5 Result

The results show that the login system successfully validates user credentials. It correctly handles errors by displaying appropriate messages for invalid or empty inputs. The system prevents unauthorized access by rejecting incorrect login attempts. Overall, the login feature functions as expected and provides a secure and reliable user authentication process.

Table 6.1 Unit Testing 1: User Login

No.	Test Case	Input	Expected Result	Result
1	Verify login with valid credentials	Correct email and Password	User Successfully logs in to user dashboard	Pass
2	Verify login with invalid credentials	Wrong email and password	Error message displayed	Pass

6.1.9 Unit Testing 2: Search Product

The purpose of this unit test is to ensure that the product search functionality of the Buy Best system works correctly and efficiently. It verifies that users are able to search for products using keywords, brand names, or specifications without any errors. The test also checks

whether the system retrieves accurate and relevant results from the database or external sources. It ensures that filtering and search processing are working properly to improve result accuracy. Additionally, it validates that the system handles invalid or empty search queries appropriately. Overall, this testing ensures that the search feature is fast, reliable, and user-friendly.

6.1.9.1 Test Objective

The objective of this test is to verify that the search feature returns correct and relevant product results based on user input. It ensures that the system correctly processes search queries and retrieves matching products from multiple sources. It also checks whether irrelevant or incorrect queries are handled properly. The test aims to confirm that the search functionality is efficient, accurate, and responsive.

6.1.9.2 Test Inputs

The test inputs include different types of search queries to evaluate system behavior. These include valid product names such as laptops, mobiles, or specific brands. Partial keywords and specifications are also used to test search flexibility. Invalid or random text inputs are used to check error handling. Empty search fields are also tested to ensure proper validation messages are displayed.

6.1.9.3 Expected Output

The expected output is that the system displays relevant product results for valid search queries. For invalid or irrelevant inputs, the system should show appropriate messages or suggestions. If the search field is empty, the system should prompt the user to enter a query. The results should be accurate, well-organized, and displayed in a structured format.

6.1.9.4 Test Procedure

The test procedure starts by entering a valid product name into the search bar. The search button is then clicked to initiate the query. The system response is observed to check if relevant products are displayed. The same steps are repeated using partial, invalid, and empty inputs. This ensures that all possible search scenarios are properly tested.

6.1.9.5 Result

The results confirm that the search functionality works correctly in all cases. The system successfully retrieves relevant products for valid queries and handles incorrect inputs

properly. It also provides appropriate messages for empty or invalid searches. Overall, the search feature is reliable, efficient, and meets the system requirements.

Table 6.2 Unit Testing 2: Search Product

No.	Test Case	Input	Expected Result	Result
1	Search for a product	Keyword: "iPhone"	Relevant product results are displayed	Pass
2	Empty search input	No keyword	Warning message shown	Pass

6.2 Functional Testing

Functional testing was conducted after unit testing to verify that each feature of our project works according to the specified requirements. It ensures that all major functionalities such as product search, filtering, comparison, AI-based recommendations, bookmarking, and saved searches operate correctly. This testing focuses on validating system behavior from the user's perspective by checking whether each feature produces the expected output. It also ensures that all modules interact properly and provide accurate results when integrated together. Different test cases are executed to cover both normal and abnormal user inputs. Overall, functional testing confirms that the system meets its intended functional requirements and provides smooth user experience.

6.2.1 Test Objective

The objective of functional testing is to ensure that every feature of our project works according to its defined requirements. It verifies that the system performs all required operations such as searching products, comparing items, and generating recommendations correctly. It also checks whether each function responds accurately to user inputs. The goal is to ensure that the system behaves as expected in all functional scenarios.

6.2.2 Test Scope

The scope of functional testing includes all major modules of the system. This includes product search, filtering system, comparison feature, AI recommendation engine, user login, bookmarking, and saved searches. It also covers system responses for valid and invalid inputs. Both frontend and backend functionalities are included in this testing phase.

6.2.3 Test Inputs

The test inputs include valid product search queries, multiple product selections for comparison, and user interaction data for recommendations. Invalid inputs such as empty fields, wrong credentials, and random text are also used. Different user actions like saving products and applying filters are tested. These inputs help ensure that the system handles all possible scenarios.

6.2.4 Expected Output

The expected output is that the system should return correct and relevant results for all valid inputs. It should display error messages for invalid or incomplete inputs. The comparison feature should show accurate side-by-side product details. AI recommendations should be relevant to user behavior and preferences. Overall, the system should respond correctly to all functional operations.

6.2.5 Test Procedure

The test procedure begins by executing each system feature individually. Users perform actions such as searching products, applying filters, comparing items, and saving products. The system response is observed for correctness and accuracy. The same steps are repeated with invalid inputs to test error handling. This ensures complete validation of all functional modules.

6.2.6 Result

The results show that all functional modules of our project are working as expected. The system successfully handles valid and invalid inputs. All features such as search, comparison, recommendations, and bookmarking function correctly. Errors are handled properly with appropriate messages. Overall, the system meets all functional requirements effectively.

6.2.7 Functional Testing 1: Login with Different Roles

Functional testing for login with different roles was performed to ensure that our project correctly identifies users and administrators and redirects them to their appropriate dashboards. This test verifies that role-based authentication is working properly and that users only access features allowed for their role. It ensures that normal users can access search, comparison, and bookmarking features, while admin users can access system management functions. The testing also checks that unauthorized access is restricted and proper validation is applied during login. Different scenarios such as valid user login, valid admin login, and invalid credentials were tested. Overall, this testing ensures secure and correct role-based access control in the system.

6.2.7.1 Test Objective

The objective of this test is to verify that the system correctly differentiates between user roles during login. It ensures that users and admins are redirected to their respective dashboards after successful authentication. It also checks that unauthorized access is prevented. The goal is to confirm proper role-based access control in the system.

6.2.7.2 Test Inputs

The test inputs include valid user credentials, valid admin credentials, and invalid login details. Different combinations of email and password are used to test role detection. Empty input fields are also tested to ensure proper validation. These inputs help verify correct system behavior for different login scenarios.

6.2.7.3 Expected Output

The expected output is that users are redirected to the user dashboard after successful login. Admin users should be redirected to the admin dashboard with additional controls and features. Invalid credentials should generate error messages. The system should prevent unauthorized access for incorrect login attempts.

6.2.7.4 Test Procedure

The test procedure begins by entering valid user login credentials and checking redirection. The same process is repeated for admin login credentials. Invalid and empty inputs are then tested to observe system responses. Each result is verified against expected behavior to ensure correctness.

6.2.7.5 Result

The results confirm that the system correctly identifies user roles during login. Users are redirected to the appropriate dashboards based on their role. Invalid login attempts are properly handled with error messages. Overall, role-based login functionality works accurately and securely.

Table 6.3 Functional Testing 1: Login with different roles

No.	Test Case	Input	Expected Result	Result
1	Login as user	User email	User dashboard	Pass
2	Login as admin	Admin email	Admin dashboard	Pass

6.1.4 Integration Testing

Integration testing is the process of checking how different modules of our project work together as a single integrated system. It ensures that individual components such as product search, comparison engine, AI recommendation module, user authentication, and database properly communicate with each other. This testing verifies that data flows correctly between frontend, backend, and external APIs without any loss or mismatch of information. It also checks that combined features produce accurate and consistent results when used together. Different module interactions are tested to identify interface issues and data transfer errors. Overall, integration testing ensures that the complete system works smoothly as one unified application.

6.1.4.1 Test Objective

The objective of integration testing is to verify that all modules of our project interact correctly with each other. It ensures that data is properly shared between different components like search, comparison, and recommendation systems. It also checks that combined functionalities work without conflicts. The goal is to ensure smooth communication between system modules.

6.1.4.2 Test Scope

The scope of integration testing includes all major system modules such as frontend interface, backend logic, database, and external APIs. It covers interactions between search engine,

product comparison module, AI recommendation system, and user management. It also includes testing of data exchange between different layers of the system. This ensures complete system integration.

6.1.4.3 Test Inputs

The test inputs include user search queries, product selection requests, comparison actions, and recommendation triggers. Data from external APIs is also used to test integration with live product sources. Invalid or incomplete inputs are used to test system robustness. These inputs help verify smooth module interaction.

6.1.4.4 Expected Output

The expected output is that all modules should work together without errors. Search results should be correctly passed on to comparison and recommendation modules. Data from APIs should be properly displayed in the system. No data loss or mismatch should occur during module interaction.

6.1.4.5 Test Procedure

The test procedure involves executing multiple system features together. A user performs a search, selects products, and compares them while the system processes AI recommendations in the background. Data flow between modules is observed for correctness. The same process is repeated with different inputs to ensure consistency.

6.1.4.6 Result

The results show that all system modules integrate properly with each other. Data flows smoothly between search, comparison, and recommendation components. No major errors or mismatches were found during testing. Overall, the system performs efficiently as a fully integrated application.

Table 6.4 Integrating Testing

No.	Test Case	Input	Expected Result	Result
1	Search and view product details	Search Product	Results shown and details page opens	Pass

2	Compare products	Select 2 products	Comparison table displayed	Pass
3	Save Search	Click save search	Search Saved successfully	Pass
4	AI recommendation	Search product	AI suggestion displayed	Pass

6.2 Automated Testing

Automated testing is the process of testing the **BuyBest** system using specialized tools and scripts instead of manual checking. It helps in quickly executing repeated test cases for features such as login, search, product comparison, and AI-based recommendations. This testing improves accuracy by reducing human errors and ensures consistent results across multiple test runs. Automated testing is especially useful for large systems where manual testing would take more time and effort. It also helps in early detection of bugs during development and integration phases. Overall, automated testing increases efficiency, reliability, and speed of the testing process.

6.2.1 Test Objective

The objective of automated testing is to verify that key functionalities of our project work correctly without manual intervention. It ensures that repeated test cases can be executed quickly and accurately. It also helps in improving testing efficiency and reducing human effort. The goal is to ensure system reliability through automated verification.

6.2.2 Test Scope

The scope of automated testing includes major system features such as user login, product search, filtering, comparison, bookmarking, and recommendations. It also covers backend validation and API responses. Regression testing is included to ensure new updates do not break existing functionality. This ensures full system coverage with automation tools.

6.2.3 Test Inputs

The test inputs include predefined test cases such as valid and invalid login credentials, product search queries, comparison selections, and filter values. Automated scripts are used to simulate user actions repeatedly. Data sets are prepared to test different scenarios. These inputs help ensure consistent system behavior.

6.2.4 Expected Output

The expected output is that all automated test cases should pass successfully without errors. The system should return correct results for valid inputs and appropriate error messages for invalid inputs. All modules should behave consistently during repeated execution. The output should match expected system behavior.

6.2.5 Test Procedure

The test procedure involves running automated test scripts using testing tools. Each module is tested by executing predefined scenarios automatically. The system responses are recorded and compared with expected results. Any mismatch is logged as an error for correction.

6.2.6 Result

The results show that our project performs correctly under automated testing. Most test cases pass successfully with expected outputs. Minor issues, if any, are identified and fixed during development. Overall, automated testing confirms system stability and efficiency.

Table 6.5 Results

Tool Name	Applied On	Results
Selenium	Login, Search, Comparison	All test cases passed successfully
Postman	API testing (AI & backend)	APIs responded correctly
PHPUnit / Laravel Testing	Backend logic	All modules working correctly

Chapter 7

7. Conclusion and Future Work

The **BuyBest** system was successfully designed and developed as an intelligent product search and comparison platform that integrates multiple e-commerce sources into a single system. It provides users with advanced features such as product search, comparison, AI-based recommendations, and saved searches in one unified interface. The system reduces the need to visit multiple websites and helps users make faster and more informed purchasing decisions. It is built using modern technologies to ensure efficiency, scalability, and smooth performance. All requirements were successfully implemented and tested during development. The system effectively achieves its goal of improving the online shopping experience for users.

7.1 Conclusion

The conclusion section summarizes the overall success and outcome of our project. It highlights how the system was developed, tested, and validated according to the defined requirements. The system provides an efficient solution for product searching and comparison with AI-based support. It ensures accurate results and improves user decision-making. Overall, the system is stable, functional, and meets all objectives.

7.1.1 System Development Success

The **BuyBest** system was successfully developed using a structured design approach. All modules including search, comparison, and recommendations were properly implemented. The development process ensured smooth integration between components. The final system is fully functional and reliable.

7.1.2 Feature Integration

All system features are integrated into a single platform for better usability. Features such as search, filtering, bookmarking, and AI recommendations work together smoothly. Data flows correctly between modules. This integration improves system efficiency.

7.1.3 User Experience

The system provides a simple and user-friendly interface. Users can easily navigate between different features without confusion. The design is responsive and works on multiple devices. This enhances overall usability.

7.1.4 Testing and Validation

The system was tested using different testing methods including unit, integration, and system testing. Errors were identified and fixed during development. All features performed correctly under test cases. This ensures system reliability.

7.1.5 Final Outcome

The final system meets all functional requirements successfully. It delivers accurate product data and comparisons. AI recommendations improve decision-making. Overall, the system performs efficiently.

7.2 Future Work

The future work section describes possible improvements and enhancements for the **BuyBest** system. It focuses on expanding system capabilities, improving AI accuracy, and increasing user accessibility. Future enhancements will make the system more advanced and user-friendly. These improvements will also increase system performance and scalability. The system has strong potential for further development in the e-commerce domain.

7.2.1 Expanded Platform Integration

More e-commerce platforms can be added to increase product availability [17]. This will provide users with more choices and better comparisons. It will improve system accuracy and coverage. The system will become more powerful.

7.2.2 Advanced AI Features

AI models can be improved for better recommendation accuracy. Machine learning can analyze deeper user behavior patterns [7]. This will provide more personalized suggestions. It will enhance decision-making quality.

7.2.3 Mobile Application Development

A mobile app can be developed for better accessibility. Users will be able to access the system anytime. It will improve convenience and usability. Mobile support will increase system reach [12].

7.2.4 Real-Time Updates

Real-time price tracking and notifications can be added. Users will receive instant alerts for price changes. This helps with better purchasing decisions. It improves user experience [10].

7.2.5 System Optimization

System performance and security can be further improved. Optimization will allow users to handle more efficiently. It will ensure fast response time. This increases system scalability.

7.3 Final Remarks

The final remarks section highlights the overall importance and impact of our project. It explains how the system successfully achieves its goal of simplifying online shopping. The system integrates modern technologies to provide complete product search and comparison solutions. It demonstrates practical implementation of AI in e-commerce. The project has strong potential for future enhancement and real-world use.

7.3.1 Project Achievement

The **BuyBest** system successfully fulfills its main objective of improving product search and comparison. It integrates multiple features into a single platform. The system provides accurate and organized information. It simplifies online shopping for users.

7.3.2 Practical Application

The system can be used in real-world e-commerce environments. It demonstrates effective use of AI and web technologies. The design is practical and scalable. It shows strong implementation value.

7.3.3 User Benefits

Users benefit from faster search and better comparison tools. It reduces time spent on multiple websites. The system improves decision-making. It enhances shopping experience [19].

7.3.4 System Strength

The system is reliable, efficient, and user-friendly. It handles multiple processes smoothly. The architecture supports scalability. It is stable and well-structured.

7.3.5 Future Potential

The system can be expanded into a full-AI-based shopping assistant. It has strong potential for growth and improvement. Additional features can increase its usability [11]. It can be widely adopted in future.

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